

MSF SITS · CLOUD INFRASTRUCTURE ENGINEER CASE STUDY

Azure Platform Lab

Demo guide: accounts, sign-in, links and evidence

Dev 10.10.0.0/16 · Test 10.20.0.0/16 · Prod 10.30.0.0/16 (own subscription)

START HERE

Accounts and sign-in

Account	Sign-in name	Access
Lab Admin	admin@peterhudcovicgmail.onmicrosoft.com	Azure portal (all environments), Azure DevOps, app in Dev / Test / Prod
Prod Admin	admin-prod@peterhudcovicgmail.onmicrosoft.com	Prod only: read-only portal, Prod Ops VM via Bastion (Entra ID), changes via PIM
opsadmin	local account on each Ops VM	Dev and Test Ops VM via Bastion (VM password); Prod emergency only
Peter Hudcovic	tenant owner	Break-glass account, not used day to day

Sign in first

1	Azure portal as Lab Admin	https://portal.azure.com
2	Azure DevOps as Lab Admin	https://dev.azure.com/peterhudcovic-lab/azure-appservice-platform-lab
3	New InPrivate window, Azure portal as Prod Admin	https://portal.azure.com
4	Source code	https://github.com/PeterHudcovic/azure-appservice-platform-lab

Where to click

Azure portal

Subscriptions	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Resources%2Fsubscriptions
Resource groups	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Resources%2FresourceGroups
Virtual machines	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Compute%2FVirtualMachines
Virtual networks	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Network%2FvirtualNetworks
Private DNS zones	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Network%2FprivateDnsZones
Application Gateways	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Network%2FapplicationGateways
App Services	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Web%2Fsites
Key Vaults	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.KeyVault%2Fvaults
Managed identities	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.ManagedIdentity%2FuserAssignedIdentities
Bastions	https://portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Network%2FbastionHosts
Prod delete lock	https://portal.azure.com/#@peterhudcovicgmail.onmicrosoft.com/resource/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f/resourceGroups/rg-sits-prod-app-swc/locks
Alerts (Azure Monitor)	https://portal.azure.com/#view/Microsoft_Azure_Monitoring/AzureMonitoringBrowseBlade/~/alertsV2

Microsoft Entra ID

Users	https://portal.azure.com/#view/Microsoft_AAD_UsersAndTenants/UserManagementMenuBlade/~/AllUsers
Groups	https://portal.azure.com/#view/Microsoft_AAD_IAM/GroupsManagementMenuBlade/~/AllGroups
Enterprise applications	https://portal.azure.com/#view/Microsoft_AAD_IAM/StartboardApplicationsMenuBlade/~/AppAppsPreview
Privileged Identity Management	https://portal.azure.com/#view/Microsoft_Azure_PIMCommon/CommonMenuBlade/~/quickStart
Conditional Access	https://portal.azure.com/#view/Microsoft_AAD_ConditionalAccess/ConditionalAccessBlade/~/Policies

Azure DevOps

Pipelines	https://dev.azure.com/peterhudcovic-lab/azure-appservice-platform-lab/_build
Environments and approvals	https://dev.azure.com/peterhudcovic-lab/azure-appservice-platform-lab/_environments
Service connections	https://dev.azure.com/peterhudcovic-lab/azure-appservice-platform-lab/_settings/adminservices
Agent pools	https://dev.azure.com/peterhudcovic-lab/azure-appservice-platform-lab/_settings/agentqueues

Application (open only from the Ops VM of the same environment)

Dev	https://app.dev.sits.internal
Test	https://app.test.sits.internal
Prod	https://app.prod.sits.internal

1 · Overview

Two subscriptions: Dev and Test in sub-app-nonprod, Prod in its own sub-app-prod.

Home > Subscriptions

Subscriptions

...



Compare the different Azure subscription types

Forecast my cost for next billing period

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Add

 Manage Policies

 View Requests

 View eligible subscriptions

 Export to CSV

 Analyze

Global administrators can manage all subscriptions in this list by updating their policy setting [here](#).

Showing subscriptions in Default Directory directory. Don't see a subscription? [Switch directories](#)

 Search for any field...

Subscriptions : **Filtered (2 of 2)**

My role == **all**

Status == **all**

 Add filter

☐ Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost	Secure Score ↑↓	Parent management group ↑↓	Status ↑↓
☐ sub-app-nonprod	8b38e1d5-139f-4006-8f2b-ffb32a6ac114	Owner	€2.39	-	Tenant Root Group	✔ Active
☐ sub-app-prod	61845030-2daa-4797-a97a-59a7fd8d478f	Owner	0.00	-	Tenant Root Group	✔ Active

Dev network resource group: ASGs, Bastion, private DNS zone, DevOps pools and NAT Gateway.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

admin@peterh
DEFAULT DIRECTORY

Home

rg-sits-dev-network-swc
Resource group

Help me generate Terraform for this resource group configuration. Generate Bicep code to duplicate this resource group. Export resource groups using Bicep or Terraform

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template Open in mobile

Overview

Activity log

Access control (IAM)

Tags

Analyze (Preview)

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name 1	Type	Location
asg-sits-dev-ops-vm-swc	Application security group	Sweden Central
asg-sits-dev-pe-kv-admin-swc	Application security group	Sweden Central
asg-sits-dev-pe-kv-app-swc	Application security group	Sweden Central
asg-sits-dev-pe-kv-cert-swc	Application security group	Sweden Central
asg-sits-dev-pe-webapp-swc	Application security group	Sweden Central
bas-sits-dev-swc	Bastion	Sweden Central
dc-sits-dev-swc	Dev center	Sweden Central
dcp-sits-dev-swc	Project	Sweden Central
dev.sits.internal	Private DNS zone	Global
mdp-sits-dev-app-swc	Managed DevOps Pool	Sweden Central
mdp-sits-dev-infra-swc	Managed DevOps Pool	Sweden Central
ng-sits-dev-swc	NAT gateway	Sweden Central

Showing 1 - 12 of 29. Display count: au... 1 2 3 New Browse Experience

Dev Ops VM with its network interface and disk.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

admin@peterhudcovic...
DEFAULT DIRECTORY (PETERHU...

Home

rg-sits-dev-network-swc
Resource group

Help me generate Terraform for this resource group configuration. Generate Bicep code to duplicate this resource group. Export resource groups using Bicep or Terraform

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template Open in mobile Group by none

Overview

Activity log Access control (IAM) Tags Analyze (Preview) Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials

Resources Recommendations

vm-sits Type equals all Location equals all Add filter

Name 1	Type	Location
nic-vm-sits-dev-ops-swc	Network Interface	Sweden Central
osdisk-vm-sits-dev-ops-swc	Disk	Sweden Central
vm-sits-dev-ops-swc	Virtual machine	Sweden Central

Showing 1 - 3 of 3. Display count: au...

New Browse Experience Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Prod network resource group in the separate production subscription.

Home > Resource Manager | Resource groups

Resource Manager | Resource groups

Default Directory

Search

CreateManage view

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Name ↑

NetworkWatcherRG

NetworkWatcherRG

rg-sits-dev-app-swc

rg-sits-dev-foundation-swc

rg-sits-dev-kv-swc

rg-sits-dev-network-swc

rg-sits-prod-app-swc

rg-sits-prod-foundation-swc

rg-sits-prod-kv-swc

rg-sits-prod-network-swc

rg-sits-test-app-swc

rg-sits-test-foundation-swc

rg-sits-test-kv-swc

rg-sits-test-network-swc

VisualStudioOnline-4D0805B4EF76

rg-sits-prod-network-swc

Resource group

Search

CreateManage viewDelete resource groupRefreshExport to CSVOpen queryAssign tags

Group by none

Overview

Activity log

Access control (IAM)

Tags

Analyze (Preview)

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Subscription (move) : sub-app-prod

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Deployments : No deployments

Location : Sweden Central

Tags (edit) : costCenter : sits-labenvironment : prodShow more (1)

ResourcesRecommendations

prod adminType equals allLocation equals allAdd filter

Name ↑

Type

Location

asg-sits-prod-pe-kv-admin-swc

Application security group

Sweden Central

Layer 0 storage: Terraform state, shared package store and flow logs per environment.

Storage accounts

★ ...

Which storage accounts have unused or inactive containers?

Analyze alerts across storage accounts

Help me identify cost-saving opportunities in storage accounts

×

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Create

↶ Restore

Manage view ▾

↺ Refresh

⬇ Export to CSV

Open query

|

Assign tags

Delete

+ Add to service group

Group by none ▾

Filter for any field...

Subscription equals all

Resource Group equals all ×

Location equals all ×

+ Add filter

☐	Name ↑		Type	Kind	Resource Group	Location	Subscription	Defender for storage status
☐	stsitsflowdevswc	...	Storage account	StorageV2	rg-sits-dev-foundation-swc	Sweden Central	sub-app-nonprod	Not protected
☐	stsitsflowprodswc	...	Storage account	StorageV2	rg-sits-prod-foundation-swc	Sweden Central	sub-app-prod	Not protected
☐	stsitsflowtestswc	...	Storage account	StorageV2	rg-sits-test-foundation-swc	Sweden Central	sub-app-nonprod	Not protected
☐	stsitspkgswc	...	Storage account	StorageV2	rg-sits-dev-foundation-swc	Sweden Central	sub-app-nonprod	Not protected
☐	stsitsfdevswc	...	Storage account	StorageV2	rg-sits-dev-foundation-swc	Sweden Central	sub-app-nonprod	Not protected
☐	stsitsfprodswc	...	Storage account	StorageV2	rg-sits-prod-foundation-swc	Sweden Central	sub-app-prod	Not protected
☐	stsitsftestswc	...	Storage account	StorageV2	rg-sits-test-foundation-swc	Sweden Central	sub-app-nonprod	Not protected

2 · CI/CD: build once, deploy many

Successful build, deploy and infrastructure pipelines for all environments.

A

azure-appservice-platf...

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Azure DevOps

peterhudcovic-lab

/

azure-appservice-platform-l...

/

Pipelines

Search

≡

🔒

🔔

⚙️

LA

New pipeline

⋮

Filter pipelines

Pipelines

RecentAllRuns

Recently run pipelines

Pipeline	Last run
✓ sits-infra-prod-l2 📁 sits	#20261005.1 • Record monitoring alerts and infrastructure pipelines in build log 12 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 2h ago 🕒 12m 52s
✓ sits-infra-prod-l1 📁 sits	#20261005.1 • Record monitoring alerts and infrastructure pipelines in build log 12 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 2h ago 🕒 15m 36s
✓ sits-infra-test-l2 📁 sits	#20261005.1 • Add testing infrastructure pipelines with pipeline identity roles and client ID variable 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 1m 53s
✓ sits-infra-test-l1 📁 sits	#20261005.1 • Add testing infrastructure pipelines with pipeline identity roles and client ID variable 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 44s
✓ sits-infra-dev-l1 📁 sits	#20261005.2 • Accept the Azure DevOps organization URL with a trailing slash in layer 1 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 43s
✓ sits-infra-dev-l2 📁 sits	#20261005.1 • Prepare development infrastructure pipelines: pipeline identity roles, client ID variable, verified apply 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 1m 45s
✓ sits-deploy-prod 📁 sits	#20261005.1 • Add the production deploy pipeline with the prod-app approval environment 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 3m 14s
✓ sits-deploy-test 📁 sits	#20261005.2 • Rely on the pipeline health check instead of the App Service start tracking during deployment 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 1m 33s
✓ sits-deploy-dev 📁 sits	#20261005.2 • Share deployment steps between environments, add sign-in check, and add the testing deploy pipeline 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 2m 3s
✓ sits-build 📁 sits	#20261005.1 • Add Azure DevOps pipelines for build, development deploy, layer 1 and layer 2 infrastructure, and destroy 👤 Manually run by LA 🛠️ c laude / dev - layer 2 🕒 Yesterday 🕒 36s

Thirteen Azure service connections, all workload identity federation, no secrets.

Service connections

Securi

Filter by keywords

Created by>Type

Name	Type	Authentication method
 build	Azure Resource Manager	Workload Identity federation (Manual)
 dev-deploy	Azure Resource Manager	Workload Identity federation (Manual)
 dev-destroy	Azure Resource Manager	Workload Identity federation (Manual)
 dev-infra-l1	Azure Resource Manager	Workload Identity federation (Manual)
 dev-infra-l2	Azure Resource Manager	Workload Identity federation (Manual)
 github-peterhudcovic	GitHub	Grant authorization
 prod-deploy	Azure Resource Manager	Workload Identity federation (Manual)
 prod-destroy	Azure Resource Manager	Workload Identity federation (Manual)
 prod-infra-l1	Azure Resource Manager	Workload Identity federation (Manual)
 prod-infra-l2	Azure Resource Manager	Workload Identity federation (Manual)
 test-deploy	Azure Resource Manager	Workload Identity federation (Manual)
 test-destroy	Azure Resource Manager	Workload Identity federation (Manual)
 test-infra-l1	Azure Resource Manager	Workload Identity federation (Manual)
 test-infra-l2	Azure Resource Manager	Workload Identity federation (Manual)

Prod deploy service connection uses OpenID Connect federation with the Prod deploy identity.

← prod-deploy

ID: a7e4916a-13e9-4679-b9a4-e282229c8988

Overview

Usage history

Approvals and checks

Details

Service connection type



Azure Resource Manager

using workload identity federation with openid connect

[Manage service connection roles](#)

[Manage Managed Identity](#)

Creator



Lab Admin

Description

Production application deployment after approval, identity id-sits-prod-deploy-swc

Workload Identity federation details

Issuer:

https://login.microsoftonline.com/3bdeb384-5dba-4624-a330-01211232fa30/v2.0

🔗

Subject identifier:

/eid1/c/pub/t/hLPeO7pdJEajMAEhEjL6MA/a/rISbSSETf0KqFyZ8ppdXmA/sc/4d0805b4-ef76-4476-9faa-183c38a42d17/a7e4916a-13e9-4679-b9a4-e282229c8988

🔗

The prod-app environment requires an approval before production deployment.

A

azure-appservice-platf...

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

←

prod-app

Production application deployment. Mandatory approval; shows the exact build run whose SHA-256 fingerprint is verified again before deployment.

Add resource

:

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel. [Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	LA

Approval settings: Lab Admin is the approver for production deployments.

Azure DevOps

peterhudcovic-lab

/ azure-appservice-platf...

/ Pipelines

/ Environments

/ prod-app

A

azure-appservice-platf...

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

←

prod-app

Production application deployment. Mandatory approval; shows the exact build run whose SHA-256 fingerprint i

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.

Order	Display name	Type
1	All approvers must approve	Approvals

Approvals

×

Approvers

Lab Admin

×

+

Instructions to approvers (optional)

Production deployment: check the build run, version, and SHA-256 fingerprint shown in the run, and the testing results, before approving

↑

↓

Advanced

^

☒

Allow approvers to approve their own runs

Control options

^

Timeout

30

Days

↓

Separate app and infra agent pools for Dev, Test and Prod.

peterhudcovic-lab / azure-appservice-platform-l... / Settings / Agent pools

Project Settings

azure-appservice-platform-lab

General

Overview

Teams

Permissions

Notifications

Service hooks

Dashboards

Boards

Project configuration

Team configuration

GitHub connections

Pipelines

Agent pools

Parallel jobs

Settings

Test management

Service connections

Agent pools

Name

Azure Pipelines

Azure Pipelines

Default

Azure Pipelines

GitHub-hosted Agents

Azure Pipelines

mdp-sits-dev-app-swc

Lab Admin

mdp-sits-dev-infra-swc

Lab Admin

mdp-sits-prod-app-swc

Lab Admin

mdp-sits-prod-infra-swc

Lab Admin

mdp-sits-test-app-swc

Lab Admin

mdp-sits-test-infra-swc

Lab Admin

Peter Hudcovic · Azure Platform Lab

16

Dev app agent pool runs inside the Dev network.

Project Settings

azure-appservice-platform-lab

General

Overview

Teams

Permissions

Notifications

Service hooks

Dashboards

Boards

Project configuration

Team configuration

GitHub connections

Pipelines

Agent pools

Parallel jobs

Settings

Test management

Service connections

XAML build services

Repos

Repositories

Migration to GitHub

Artifacts

Storage

Test

cloud

mdp-sits-dev-app-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

LA

Lab Admin

×

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-dev-app-swc

This Pool is managed via the azure resource: /subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114/resourceGroups/rg-sits-dev-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-dev-app-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

• No contacts provided

Pool Details

• Region: swedencentral

• SKU: Standard_D2ads_v5

• Parallelism: 1

• Stateful: False

• Predictive Provisioning Mode: None

• Vnet State: Subnet Delegated

• Public IPs: 0

Peter Hudcovic · Azure Platform Lab

17

Dev infra agent pool runs inside the Dev network.

 mdp-sits-dev-infra-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

LA

Lab Admin

×

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-dev-infra-swc

This Pool is managed via the azure resource: /subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114/resourceGroups/rg-sits-dev-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-dev-infra-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

- No contacts provided

Pool Details

- Region: swedencentral
- SKU: Standard_D2ads_v5
- Parallelism: 1
- Stateful: False
- Predictive Provisioning Mode: None
- Vnet State: Subnet Delegated
- Public IPs: 0

Test app agent pool runs inside the Test network.

 mdp-sits-test-app-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

LA

Lab Admin

✕

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-test-app-swc

This Pool is managed via the azure resource: /subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114/resourceGroups/rg-sits-test-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-test-app-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

- No contacts provided

Pool Details

- Region: swedencentral
- SKU: Standard_D2as_v5
- Parallelism: 1
- Stateful: False
- Predictive Provisioning Mode: None
- Vnet State: Subnet Delegated
- Public IPs: 0

Test infra agent pool runs inside the Test network.

 mdp-sits-test-infra-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

LA

Lab Admin

×

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-test-infra-swc

This Pool is managed via the azure resource: /subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114/resourceGroups/rg-sits-test-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-test-infra-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

- No contacts provided

Pool Details

- Region: swedencentral
- SKU: Standard_D2as_v5
- Parallelism: 1
- Stateful: False
- Predictive Provisioning Mode: None
- Vnet State: Subnet Delegated
- Public IPs: 0

Prod app agent pool runs inside the Prod network.

 mdp-sits-prod-app-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

 Lab Admin

✕

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-prod-app-swc

This Pool is managed via the azure resource: /subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f/resourceGroups/rg-sits-prod-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-prod-app-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

- No contacts provided

Pool Details

- Region: swedencentral
- SKU: Standard_D2ads_v5
- Parallelism: 1
- Stateful: False
- Predictive Provisioning Mode: None
- Vnet State: Subnet Delegated
- Public IPs: 0

Prod infra agent pool runs inside the Prod network.

 mdp-sits-prod-infra-swc

Jobs

Details

Security

Approvals and checks

Analytics

Owner

Undo

Save

LA

Lab Admin

×

Description

Edit

Undo

Save

Managed Devops Pool: mdp-sits-prod-infra-swc

This Pool is managed via the azure resource: /subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f/resourceGroups/rg-sits-prod-network-swc/providers/Microsoft.DevOpsInfrastructure/pools/mdp-sits-prod-infra-swc. Please do not modify or delete this pool directly in Azure DevOps.

Read more about Managed Devops Pools in our [documentation](#)

If you have any questions or need to request changes to this pool, please contact the admins in the Security section or one of the following contacts:

- No contacts provided

Pool Details

- Region: swedencentral
- SKU: Standard_D2ads_v5
- Parallelism: 1
- Stateful: False
- Predictive Provisioning Mode: None
- Vnet State: Subnet Delegated
- Public IPs: 0

3 · Users, groups and application sign-in

One app-user group per environment controls who may sign in to the application.

Home > Default Directory | Groups > Groups

 **Groups** | All groups

Default Directory

◇

«

 New group

 Download groups

 Refresh

 Manage view ▾

 Delete

 Got feedback?

 Overview

 **All groups**

 Insights (Preview)

 Deleted groups

 Diagnose and solve problems

> Settings

> Activity

> Help

 [Microsoft Entra has a simpler, integrated experience for managing all your Identity and Access Management needs. Try the new Microsoft Entra admin center! !\[\]\(89ed16af53c63062d58e67cd60de5b38_img.jpg\)](#)

 Search

 Add filter

Search mode

Contains

3 groups found

☐	Name ↑	Object Id	Group type	Membership type	Email
☐	 grp-sits-dev-app-users	8deaa12c-efa7-406e-a36a-b7f61636e3b0	Security	Assigned	
☐	 grp-sits-prod-app-users	c9c1cbf5-0300-4e99-b9aa-5ceb46ec3a69	Security	Assigned	
☐	 grp-sits-test-app-users	71ef8af0-6d89-4b2c-b4bd-186bf6b10bf8	Security	Assigned	

Peter Hudcovic · Azure Platform Lab

24

Dev app-user group membership.

Home > Default Directory | Groups > Groups | All groups > grp-sits-dev-app-users

 **grp-sits-dev-app-users** | Members

Group

+

Add members

 Bulk operations

 Refresh

 Manage view

×

Remove

 Got feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Help

Direct members

All members

Search

Add filter

1 group member found

	Name ↑	Type	Email
☐	LA Lab Admin	User	

Peter Hudcovic · Azure Platform Lab

25

Test app-user group membership.

**grp-sits-test-app-users** | Members

Group

✕

«

+

Add members

📄

Bulk operations

↺

Refresh

⚙️

Manage view

✕

Remove

💬

Got feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Help

Direct members

All members

🔍

Search

🔼

Add filter

1 group member found

☐	Name ↑	Type	Email
☐	LA Lab Admin	User	

Prod app-user group membership.

**grp-sits-prod-app-users** | Members ...
Group

Overview

Diagnose and solve problems

Manage

- Properties
- Members**
- Owners
- Roles and administrators
- Administrative units
- Group memberships
- Applications
- Licenses
- Azure role assignments

Activity

Help

+ Add members

 Bulk operations ▾

 Refresh

 Manage view ▾

×

 Remove

 Got feedback?

Direct members

All members

 Search

 Add filter

1 group member found

☐	Name ↑	Type	Email
☐	 LA Lab Admin	User	

Dev enterprise application requires assignment for sign-in.

app-sits-dev | Properties

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

Security

Conditional Access

Permissions

Token encryption

Activity

Sign-in logs

Usage & insights

Audit logs

Provisioning logs

Access reviews

Help

Support + Troubleshooting

« Got feedback?

View and manage application settings for your organization. Editing properties like display information, user sign-in settings, and user visibility settings requires Global Administrator, Cloud Application Administrator, Application Administrator roles. [Learn more.](#)

If this application resides in your tenant, you can manage additional properties on the [application registration](#).

Activation status ⓘ

✔ Activated

Enabled for users to sign-in? ⓘ

Yes No

Name * ⓘ

app-sits-dev ✔

Homepage URL ⓘ

Logo ⓘ

A

Select a file

Application ID ⓘ

f4169317-e014-44b4-8565-6aa0f818c3d8

Object ID ⓘ

b08a6f9f-f9bf-4dab-bf93-b40d38f46356

Assignment required? ⓘ

Yes No

Visible to users? ⓘ

Yes No

Notes ⓘ

Test enterprise application requires assignment for sign-in.

Home > Enterprise applications | All applications > app-sits-test

app-sits-test | Properties

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

Security

Conditional Access

Permissions

Token encryption

Activity

Sign-in logs

Usage & insights

Audit logs

Provisioning logs

Access reviews

Help

Support + Troubleshooting

<< Got feedback?

View and manage application settings for your organization. Editing properties like display information, user sign-in settings, and user visibility settings requires Global Administrator, Cloud Application Administrator, Application Administrator roles. [Learn more.](#)

If this application resides in your tenant, you can manage additional properties on the [application registration](#).

Activation status ⓘ

Activated

Enabled for users to sign-in? ⓘ

Yes No

Name * ⓘ

app-sits-test

Homepage URL ⓘ

Logo ⓘ

A

Select a file

Application ID ⓘ

5862e35e-c595-4118-80a4-15008d3b2d40

Object ID ⓘ

a82cd7f5-acef-40a9-8136-e39bb98213a1

Assignment required? ⓘ

Yes No

Visible to users? ⓘ

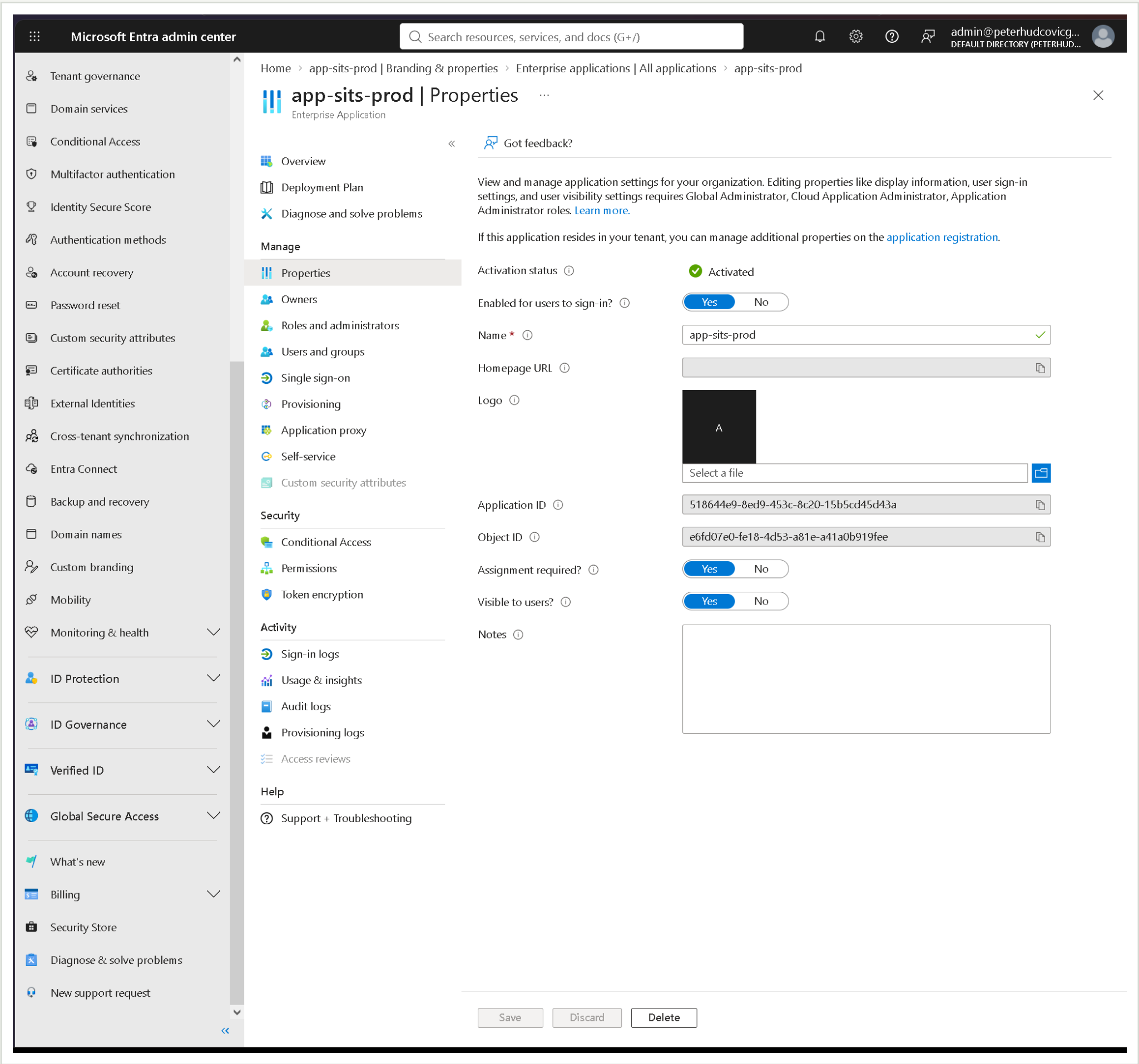
Yes No

Notes ⓘ

Peter Hudcovic · Azure Platform Lab

29

Prod enterprise application requires assignment for sign-in.



Dev application is assigned to the Dev app-user group.

...

>

Enterprise applications | All applications

>

app-sits-prod | Properties

>

Enterprise applications | All applications

>

app-sits-dev

 **app-sits-dev** | Users and groups

Enterprise Application

...

«

 Add user/group

 Edit assignment

 Remove assignment

 Update credential

 Refresh

...

 Overview

 Deployment Plan

 Diagnose and solve problems

Manage

 Properties

 Owners

 Roles and administrators

 Users and groups

 Single sign-on

 Provisioning

 Application proxy

 Self-service

 Custom security attributes

 The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

 First 200 shown, search all users & groups

	Display name	Object type
☐	LA Lab Admin	User
☐	G grp-sits-dev-app-users	Group

Test application is assigned to the Test app-user group.

Home > Enterprise applications | All applications > app-sits-test

 **app-sits-test** | Users and groups

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

<<

+ Add user/group

✎ Edit assignment

🗑 Remove assignment

🔑 Update credential

🔄 Refresh

⋮

📘

The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

🔍

First 200 shown, search all users & groups

	Display name	Object type
☐	G grp-sits-test-app-users	Group

Peter Hudcovic · Azure Platform Lab

32

Prod application is assigned to the Prod app-user group.

Home > app-sits-prod | Branding & properties > Enterprise applications | All applications > app-sits-prod

 **app-sits-prod | Users and groups** ...

Enterprise Application

«

 Add user/group

 Edit assignment

 Remove assignment

 Update credential

 Refresh

...

 Overview

 Deployment Plan

 Diagnose and solve problems

Manage

 Properties

 Owners

 Roles and administrators

 Users and groups

 Single sign-on

 Provisioning

 Application proxy

 Self-service

 Custom security attributes

 The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

 First 200 shown, search all users & groups

	Display name	Object type
☐	 Lab Admin	User
☐	 grp-sits-prod-app-users	Group

Peter Hudcovic · Azure Platform Lab

33

Prod enterprise application is enabled for user sign-in.

Home > app-sits-prod | Branding & properties > Enterprise applications | All applications > app-sits-prod

app-sits-prod | Properties

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

Provisioning

Application proxy

Self-service

Custom security attributes

« Got feedback?

View and manage application settings for your organization. Editing properties like display information, user sign-in settings, and user visibility settings requires Global Administrator, Cloud Application Administrator, Application Administrator roles. [Learn more.](#)

If this application resides in your tenant, you can manage additional properties on the [application registration](#).

Activation status ⓘ

✔ Activated

Enabled for users to sign-in? ⓘ

Yes

No

Name * ⓘ

app-sits-prod ✔

Homepage URL ⓘ

Logo ⓘ

A

Select a file

Peter Hudcovic · Azure Platform Lab

34

Prod sign-in application and the app managed identity are separate identities.

Home > app-sits-prod | Branding & properties > Enterprise applications

Enterprise applications | All applications

Default Directory

Overview

Overview

Diagnose and solve problems

Manage

All applications

Private Network connectors

User settings

App launchers

Custom authentication extensions

Security

Conditional Access

Consent and permissions

<< + New application ↺ Refresh ⬇ Download (Export) ⓘ Preview info ≡ Columns ⚙ Preview features 👤 Got feedback?

View, filter, and search applications in your organization that are set up to use your Microsoft Entra tenant as their Identity Provider.

The list of applications that are maintained by your organization are in [application registrations](#).

🔍 app-sits-prod ×

+ Add filters

2 applications found

Name	↕	Object ID	Application ID	Homepage URL	Created on	↕	Activation status
A app-sits-prod		e6fd07e0-fe18-4d53-a81e-a41a...	518644e9-8ed9-453c-8c20-15b5cd...		10/5/2026	✔	Activated
I id-sits-prod-app-swc		d81bcf1e-6b04-4799-b471-62e...	05ee7ee8-83b1-404a-9e36-ed3024...		10/5/2026	✔	Activated

Peter Hudcovic · Azure Platform Lab

35

4 · Privileged access: Prod Admin and PIM

Prod Admin: Reader always active, Contributor only eligible through PIM.

Home > sub-app-prod

sub-app-prod | Access control (IAM)

Subscription

Search

⌵ ⌵

+ Add ⌵

↓ Download role assignments

≡ Edit columns

🔄 Refresh

| 🗑 Delete

| 👤 Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Security

Resource visualizer

Events

Resource groups

Resources

> Cost Management

> Billing

> Settings

> Monitoring

> Help

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription ⓘ

Privileged ⓘ

40

5000

3

View assignments

prod admin

×

Type : All

Role : All

Scope : All scopes

State : All

End time : All

Group by : Role

All (2)

Job function roles (1)

Privileged administrator roles (1)

☐	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	State ↑↓	End time ↑↓	Condition ↑↓
⌵	Contributor (1)						
☐	PA Prod Admin 0ba401cf-650b-4548-9360-bf5...	User	Contributor	🔑 This resource	Eligible Permanent	Permanent	None
⌵	Reader (1)						
☐	PA Prod Admin 0ba401cf-650b-4548-9360-bf5...	User	Reader	🔑 This resource	Active Permanent	Permanent	None

Showing 1 - 2 of 2 results.

Prod Admin can always sign in to the Prod VM as a normal user.

Home > Resource Manager | Resource groups > rg-sits-prod-network-swc

Resource Manager | Resource groups

Default Directory

Search

+ Create

Manage view

...

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Name ↑

NetworkWatcherRG

NetworkWatcherRG

rg-sits-dev-app-swc

rg-sits-dev-foundation-swc

rg-sits-dev-kv-swc

rg-sits-dev-network-swc

rg-sits-prod-app-swc

rg-sits-prod-foundation-swc

rg-sits-prod-kv-swc

rg-sits-prod-network-swc

rg-sits-test-app-swc

rg-sits-test-foundation-swc

rg-sits-test-kv-swc

rg-sits-test-network-swc

VisualStudioOnline-4D0805B4EF76

rg-sits-prod-network-swc | Access control (IAM)

Resource group

Search

+ Add

Download role assignments

Edit columns

Refresh

Delete

Feedback

Overview

Activity log

Access control (IAM)

Tags

Analyze (Preview)

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription

40

5000

Privileged

5

View assignments

Prod Admin

Type : All

Role : All

Scope : All scopes

State : All

End time : All

Group by : Role

All (3)

Job function roles (2)

Privileged administrator roles (1)

Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	State ↑↓	End time ↑↓	Condition ↑↓
Contributor (1)						
PA Prod Admin 0ba401cf-650b...	User	Contributor	Subscription (Inh...	Eligible Permanent	Permanent	None
Reader (1)						
PA Prod Admin 0ba401cf-650b...	User	Reader	Subscription (Inh...	Active Permanent	Permanent	None
Virtual Machine User Login (1)						
PA Prod Admin 0ba401cf-650b...	User	Virtual Machine User L...	This resource	Active Permanent	Permanent	None

Showing 1 - 3 of 3 results.

PIM: Contributor on the Prod subscription is eligible, not active.

Home > Privileged Identity Management | Azure resources > sub-app-prod

 **sub-app-prod | Assignments** ...

Privileged Identity Management | Azure resources

«

[+ Add assignments](#)

[☰ Review](#)

[⚙ Settings](#)

[🔄 Refresh](#)

[⬇ Export](#)

| [🗨 Got feedback?](#)

Overview

Tasks

My roles

Pending requests

Approve requests

Review access

Manage

Roles

Assignments

Alerts

Access reviews

Eligible assignments

Active assignments

Expired assignments

🔍 Search by member name

Name	Principal name	Type	Membership	Condition	Start time	End time
Contributor						
Prod Admin	admin-prod@peterhudcovicgmail.onmicrosoft.com	User	Direct	None	10/5/2026, 8:42:00 PM	Permanent

Eligible Contributor is inherited by the Prod foundation resource group.

Home > Browse > rg-sits-prod-foundation-swc

 **rg-sits-prod-foundation-swc** | Assignments

Privileged Identity Management | Azure resources

«

[+ Add assignments](#) [☰ Review](#) [⚙ Settings](#) [🔄 Refresh](#) [⬇ Export](#) | [🗣 Got feedback?](#)

Overview

Tasks

My roles

Pending requests

Approve requests

Manage

Roles

Assignments

Eligible assignments

Active assignments

Expired assignments

🔍 Search by member name

Name	Principal name	Type	Membership	Condition	Start time	End time	Action
Contributor							
Prod Admin	admin-prod@peterhudcc	User	Inherited	None	10/5/2026, 8:42:00 PM	Permanent	Remove Update

PIM: Key Vault Secrets User on the admin vault is eligible only.

Home > Browse > kv-sits-prod-adm-swc

 kv-sits-prod-adm-swc | Assignments

Privileged Identity Management | Azure resources

Overview

Tasks

My roles

Pending requests

Approve requests

Manage

Roles

Assignments

Settings

Activity

Resource audit

Activity log

My audit

«

+ Add assignments

☰ Review

⚙ Settings

🔄 Refresh

⬇ Export

|

🗨 Got feedback?

Eligible assignments

Active assignments

Expired assignments

🔍 Search by member name

Name	Principal name	Type	Membership	Condition	Start time	End time	Action
Contributor							
Prod Admin	admin-prod@peterhudco	User	Inherited	None	10/5/2026, 8:42:00 PM	Permanent	Remove Update
Key Vault Secrets User							
Prod Admin	admin-prod@peterhudco	User	Direct	None	10/5/2026, 8:41:52 PM	Permanent	Remove Update

PIM: Key Vault Secrets Officer on the app vaults is eligible only.

Home > Browse > rg-sits-prod-kv-swc

 **rg-sits-prod-kv-swc | Assignments**

Privileged Identity Management | Azure resources

Overview

Tasks

Manage

Activity

«

+ Add assignments

≡ Review

⚙ Settings

🔄 Refresh

⬇ Export

|

🗨 Got feedback?

Eligible assignments

Active assignments

Expired assignments

🔍 Search by member name

Name	Principal name	Type	Membership	Condition	Start time	End time	Action
Contributor							
Prod Admin	admin-prod@peterhudcovicgmail.onmic	User	Inherited	None	10/5/2026, 8:42:00 PM	Permanent	Remove Update
Key Vault Secrets Officer							
Prod Admin	admin-prod@peterhudcovicgmail.onmic	User	Direct	None	10/5/2026, 8:42:13 PM	Permanent	Remove Update

<

>

PIM activation: maximum 2 hours, MFA and justification required.

Home > Privileged Identity Management | Azure resources > sub-app-prod | Roles > Contributor

Role setting details - Contributor

Privileged Identity Management | Azure resources

Edit

Activation

Setting	State
Activation maximum duration (hours)	2 hour(s)
On activation, require	Azure MFA
Require justification on activation	Yes
Require ticket information on activation	No
Require approval to activate	No
Approvers	None
Require pre-approval custom extension to activate (Preview)	No
Require post-approval custom extension to activate (Preview)	No

Assignment

Setting	State
Allow permanent eligible assignment	Yes
Expire eligible assignments after	-
Allow permanent active assignment	No
Expire active assignments after	6 month(s)
Require Azure Multi-Factor Authentication on active assignment	No
Require justification on active assignment	Yes

Send notifications when members are assigned as eligible to this role:

Type	Default recipients	Additional recipients	Critical emails only
Role assignment alert	Admin	None	False
Notification to the assigned user (assignee)	Assignee	None	False
Request to approve a role assignment renewal/extension	Approver	None	False

Send notifications when members are assigned as active to this role:

Type	Default recipients	Additional recipients	Critical emails only
Role assignment alert	Admin	None	False
Notification to the assigned user (assignee)	Assignee	None	False
Request to approve a role assignment renewal/extension	Approver	None	False

Send notifications when eligible members activate this role:

Type	Default recipients	Additional recipients	Critical emails only
Role activation alert	Admin	None	False

5 · Conditional Access (report-only)

Two Conditional Access policies protect the Prod admin, in report-only mode.

Home > Enterprise applications | All applications > app-sits-dev | Properties > Conditional Access

Conditional Access | Policies

Microsoft Entra ID

Overview

Policies

Deleted Policies

Phased rollout

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

New policy

New policy from template

Upload policy file

What if

Refresh

Preview features

Got feedback?

Microsoft Entra Conditional Access policies are used to apply access controls to keep your organization secure. [Learn more](#)

Microsoft-managed policies

Policies created 0

User created policies

Policies created 2

Search policies

Add filter

2 out of 2 policies found

Policy name	Created by	State	Alert	Creation date	Modified date
SITS production: block device code sign-in for the production	USER	Report-only		10/5/2026, 8:15:20...	
SITS production: block the production admin from developm	USER	Report-only		10/5/2026, 8:37:45...	

Peter Hudcovic · Azure Platform Lab

45

Policy blocks device-code sign-in for the Prod admin.

Home > Enterprise applications | All apps

Conditional Access

Microsoft Entra ID

Overview

Policies

Deleted Policies

Phased rollout

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

Policy details

SITS production: block device code sign-in for the production admin

View or Edit

Duplicate

Download JSON

Delete

Policy details

Policy impact

State

Report-only (policy is evaluated but not enforced)

Created by

User

Users, agents or workload identities

1 user

Excluded identities

1 user, 0 groups, 0 roles

Included resources

All resources

Network

Not configured

Requirements for access

Block access

Client apps

1 included

Included authentication flows

deviceCodeFlow

Created date

10/5/26, 8:15:20 PM

Policy blocks the Prod admin from the Dev and Test networks.

Home > Conditional Access

Conditional Access

Microsoft Entra ID

Overview

Policies

Deleted Policies

Phased rollout

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

Policy details

SITS production: block the production admin from development and testing NAT addresses

View or Edit

Duplicate

Download JSON

Delete

Policy details

Policy impact

State

Report-only (policy is evaluated but not enforced)

Created by

User

Users, agents or workload identities

1 user

Excluded identities

1 user, 0 groups, 0 roles

Included resources

All resources

Network

Not configured

Requirements for access

Block access

Locations

1 included

Client apps

1 included

Created date

10/5/26, 8:37:45 PM

Named location holds the Dev and Test NAT outbound addresses.

Home > Conditional Access

Conditional Access | Named locations

Microsoft Entra ID

Overview

Policies

Deleted Policies

Phased rollout

Insights and reporting

Diagnose and solve problems

Manage

Named locations

Custom controls (Preview)

Terms of use

VPN connectivity

Authentication contexts

Authentication strengths

Classic policies

Monitoring

Sign-in logs

Audit logs

Troubleshooting + Support

New support request

<<

+ Countries location

+ IP ranges location

Named locations are used by Microsoft Entra ID to identify locations that are marked Trusted or Untrusted. Named Locations that are marked Trusted or Untrusted are used by Conditional Access policies to determine if a user is located in a trusted or untrusted location.

All Named Locations

Deleted Named Locations

Search by name

1 named location found

Name	Location type
SITS development and testing NAT addresses	IP ranges

<

Update location (IP ranges)

↑ Upload

↓ Download

🗑 Delete

Configure named location IPv4 and IPv6 ranges. [Learn more](#)

Name *

SITS development and testing NAT addresses

☐ Mark as trusted location

+

2 IP ranges found

20.240.216.144/32	🗑
9.223.113.24/32	🗑

6 · Managed identities and least privilege

Separate managed identities for app, gateway, deploy, destroy and infra layers.

Home

Managed Identities

☆ ...

List Managed Identities modified in the last 30 days using ARG.

How many Managed Identities do I have in total?

Filter Managed Identities by resource group and location.

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Create

⚙️ Manage view

↻ Refresh

📄 Export to CSV

🔗 Open query

|

🏷️ Assign tags

🗑️ Delete

+ Add to service group

≡ Group by none

🔍 id-sit

✕

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

☐	Name ↑		Type	Resource Group	Location	Subscription ID
☐	id-sits-build-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-agw-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-app-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-deploy-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-destroy-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-infra-l1-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-dev-infra-l2-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-prod-agw-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-prod-app-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-prod-deploy-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-prod-destroy-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-prod-infra-l1-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-prod-infra-l2-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59...
☐	id-sits-test-agw-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-test-app-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-test-deploy-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-test-destroy-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-test-infra-l1-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...
☐	id-sits-test-infra-l2-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb...

Prod identities live in the separate production subscription.

Managed Identities

Default Directory (peterhudcovicgmail.onmicrosoft.com)

Identify non-compliant Managed Identities using ARG.

How many Managed Identities do I have in total?

Filter Managed Identities by resource group and location.

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Delete

Add to service group

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

☐	Name	Type	Resource Group	Location	Subscription ID ↑	
☐	 id-sits-prod-agw-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-prod-app-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-prod-deploy-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-prod-destroy-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-prod-infra-l1-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-prod-infra-l2-swc	...	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f
☐	 id-sits-build-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-agw-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-app-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-deploy-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-destroy-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-infra-l1-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-dev-infra-l2-swc	...	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-agw-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-app-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-deploy-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-destroy-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-infra-l1-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114
☐	 id-sits-test-infra-l2-swc	...	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114

Build identity can write packages to the shared package storage.

Home > Managed Identities > id-sits-build-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

⚙️ Manage view

⌵

⋮

☐

Name ↑

☐

 id-sits-build-swc

 id-sits-build-swc | Azure role assignments

☆ ⋮

×

Managed Identity

🔍 Search

◇ ⌵

+ Add role assignment (Preview)

🔄 Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

⌵ Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription ^{*}

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To	Condition
Storage Blob Data Contributor	 stsitspkgswc/default/packages	Microsoft.Storage/storageAccounts/bl...	id-sits-build-swc	None

Peter Hudcovic · Azure Platform Lab

52

Build identity trusts Azure DevOps through a federated credential.

Home > Managed Identities > id-sits-build-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

⚙️ Manage view

⌵

⋮

Name ↑

🔑 id-sits-build-swc

🔑 id-sits-build-swc | Federated credentials

☆

⋮

Managed Identity

🔍 Search

⌵

⌵

🔑 Overview

📅 Activity log

👤 Access control (IAM)

🏷️ Tags

🔑 Azure role assignments

🏗️ Associated resources (preview)

🔗 Resource visualizer

⌵ Settings

🔑 Federated credentials

⌵ Properties

🔒 Locks

➤ Monitoring

➤ Automation

➤ Help

Federated credentials

Allow other identities to impersonate this managed identity by establishing a trust with an external OpenID Connect (OIDC) identity provider. This federation allows you to get tokens to access Microsoft Entra ID protected resources that this managed identity has access to. [Learn more](#)

+ Add Credential

1 of 20 configured

Name	Issuer	Subject Identifier	Delete
azure-devops-build	https://login.microsoftonline.com/3bdeb384-5dba-4624-a330-01211232fa30/v2.0	/eid1/c/pub/t/hLPeO7pdJEajMAEhEjL...	🗑️

Federated credential details: issuer and subject, no secret.

Home > Managed Identities > id-sits-build-swc | Federated credentials

Edit Federated Credential ...

Configure an identity from an external OpenID Connect Provider to get tokens as this managed identity to access Microsoft Entra ID protected services. [Learn more](#)

Federated credential scenario * ⓘ

Configure an identity managed by an external OpenID Connect Provider to access Azure resour... ▾

ⓘ

To ensure secure FIC access, make sure your issuer uses immutable subject identifiers. If you use GitLab, review the recommended configuration. [Learn more](#)

Connect your account

Enter the details of the account that you want to connect with Microsoft Entra ID. These values will be used by Microsoft Entra ID to validate the connection.

Issuer URL * ⓘ

https://login.microsoftonline.com/3bdeb384-5dba-4624-a330-01211232fa30/v2.0

Subject identifier * ⓘ

/eid1/c/pub/t/hLPeO7pdJEajMAEhEjL6MA/a/rISbSSETf0KqFyZ8ppdXmA/sc/4d0805b4-ef76-4476-...

Credential details

Enter and review the details for this credential. The credential name cannot be edited after creation.

Name * ⓘ

azure-devops-build

Audience * ⓘ

api://AzureADTokenExchange

Edit (optional)

Peter Hudcovic · Azure Platform Lab

54

Prod deploy identity has its own federated credential.

Home > Managed Identities > id-sits-prod-deploy-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

⚙️ Manage view

⌵

⋮

Name ↑

🔑 id-sits-prod-deploy-swc

🔑 id-sits-prod-deploy-swc | Federated credentials

Managed Identity

🔍 Search

⌵

⌵

🔑 Overview

📅 Activity log

👤 Access control (IAM)

🏷️ Tags

🔑 Azure role assignments

📊 Associated resources (preview)

🔗 Resource visualizer

⌵ Settings

🔑 Federated credentials

☆

⋮ Properties

🔒 Locks

> Monitoring

> Automation

> Help

Federated credentials

Allow other identities to impersonate this managed identity by establishing a trust with an external OpenID Connect (OIDC) identity provider. This federation allows you to get tokens to access Microsoft Entra ID protected resources that this managed identity has access to. [Learn more](#)

+ Add Credential

1 of 20 configured

Name	Issuer	Subject Identifier	Delete
azure-devops-prod-deploy	https://login.microsoftonline.com/3bdeb384-5dba-4624-a330-01211232fa30/v2.0	/eid1/c/pub/t/hLPeO7pdJEajMAEhEjL...	🗑️

Prod deploy identity can only read packages.

Home > Managed Identities > id-sits-prod-deploy-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

Manage view

...

Name ↑

id-sits-prod-deploy-swc

id-sits-prod-deploy-swc | Azure role assignments

Managed Identity

Search

◊ <<

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *
sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To	Condition
Storage Blob Data Reader	 stsitspkgswc/default/packages	Microsoft.Storage/storageAccounts/bl...	id-sits-prod-deploy-swc	None

Peter Hudcovic · Azure Platform Lab

56

Prod deploy identity can only deploy to the Prod app resource group.

Home > Managed Identities > id-sits-prod-deploy-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

Manage view

...

Name ↑

id-sits-prod-deploy-swc

id-sits-prod-deploy-swc | Azure role assignments

Managed Identity

Search

◊

«

+ Add role assignment (Preview)

Refresh

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *
sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
Website Contributor	rg-sits-prod-app-swc	Resource Group	id-sits-prod-deploy-swc	None

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

Dev app identity can read secrets in the Dev app vault.

 **id-sits-dev-app-swc** | Azure role assignments

Managed Identity

Search

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

- Federated credentials
- Properties
- Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Key Vault Secrets User	 rg-sits-dev-kv-swc	Resource Group	id-sits-dev-app-swc

Test app identity can read secrets in the Test app vault.

 id-sits-test-app-swc | Azure role assignments

Managed Identity

Search

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

- Federated credentials
- Properties
- Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *
sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Key Vault Secrets User	 rg-sits-test-kv-swc	Resource Group	id-sits-test-app-swc

Prod app identity can read secrets in the Prod app vault.

Home > Managed Identities > id-sits-prod-app-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

Manage view

...

Name ↑

id-sits-prod-app-swc

id-sits-prod-app-swc | Azure role assignments

Managed Identity

Search

◊

◀

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
Key Vault Secrets User	 rg-sits-prod-kv-swc	Resource Group	id-sits-prod-app-swc	None

Peter Hudcovic · Azure Platform Lab

60

Dev gateway identity can read its certificate vault.

 **id-sits-dev-agw-swc** | Azure role assignments

Managed Identity

Search

◇

«

+ Add role assignment (Preview)

↺ Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Key Vault Secrets User	 kv-sits-dev-cert-swc	Key vault	id-sits-dev-agw-swc

Test gateway identity can read its certificate vault.

id-sits-test-agw-swc | Azure role assignments

Managed Identity

Search

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

- Federated credentials
- Properties
- Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Key Vault Secrets User	kv-sits-test-cert-swc	Key vault	id-sits-test-agw-swc

Prod gateway identity can read only the Prod certificate vault.

Managed Ident...

It Directory (peterhudcovicgmail.onmicrosoft.c...

CreateManage view

Name ↑

id-sits-prod-agw-swc

id-sits-prod-agw-swc | Azure role assignments

Managed Identity

Search

OverviewActivity logAccess control (IAM)TagsAzure role assignmentsAssociated resources (preview)Resource visualizerSettingsFederated credentialsPropertiesLocksMonitoringAutomationHelp

+ Add role assignment (Preview)Refresh

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
Key Vault Secrets User	 kv-sits-prod-cert-swc	Key vault	id-sits-prod-agw-swc	None

Dev deploy identity roles.

id-sits-dev-deploy-swc | Azure role assignments

Managed Identity

Search

+ Add role assignment (Preview)

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

- Federated credentials
- Properties
- Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *
sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To	Condition
Storage Blob Data Reader	 stsitspkgswc/default/packages	Microsoft.Storage/storageAccounts/bl...	id-sits-dev-deploy-swc	None
Website Contributor	 rg-sits-dev-app-swc	Resource Group	id-sits-dev-deploy-swc	None

Test deploy identity roles.

 **id-sits-test-deploy-swc** | Azure role assignments

Managed Identity

Search

◇

«

+ Add role assignment (Preview)

↻ Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

▼ Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To	Condition
Website Contributor	 rg-sits-test-app-swc	Resource Group	id-sits-test-deploy-swc	None
Storage Blob Data Reader	 stsitspkgswc/default/packages	Microsoft.Storage/storageAccounts/bl...	id-sits-test-deploy-swc	None

Dev destroy identity roles.

**id-sits-dev-destroy-swc** | Azure role assignments

Managed Identity

Search

◊

«

+ Add role assignment (Preview)

🔄 Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Storage Blob Data Contributor	 stsitstfdevswc/default/tfstate-layer2	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-dev-destroy-swc
Storage Blob Data Contributor	 stsitstfdevswc/default/tfstate-layer1	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-dev-destroy-swc
Contributor	 rg-sits-dev-app-swc	Resource Group	id-sits-dev-destroy-swc
Contributor	 rg-sits-dev-kv-swc	Resource Group	id-sits-dev-destroy-swc
Contributor	 rg-sits-dev-network-swc	Resource Group	id-sits-dev-destroy-swc
Network Contributor	 NetworkWatcherRG	Resource Group	id-sits-dev-destroy-swc
Reader	 rg-sits-dev-foundation-swc	Resource Group	id-sits-dev-destroy-swc

Test destroy identity roles.

**id-sits-test-destroy-swc** | Azure role assignments

Managed Identity

 Add role assignment (Preview)

 Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

 Settings

 Federated credentials

 Properties

 Locks

 Monitoring

 Automation

 Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription 
sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Contributor	 rg-sits-test-network-swc	Resource Group	id-sits-test-destroy-swc
Contributor	 rg-sits-test-kv-swc	Resource Group	id-sits-test-destroy-swc
Contributor	 rg-sits-test-app-swc	Resource Group	id-sits-test-destroy-swc
Storage Blob Data Contributor	 stsitstftestswc/default/tfstate-layer2	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-test-destroy-swc
Storage Blob Data Contributor	 stsitstftestswc/default/tfstate-layer1	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-test-destroy-swc
Network Contributor	 NetworkWatcherRG	Resource Group	id-sits-test-destroy-swc
Reader	 rg-sits-test-foundation-swc	Resource Group	id-sits-test-destroy-swc
Reader	 appi-sits-dev-swc	Application Insights	id-sits-test-destroy-swc
Reader	 log-sits-dev-swc	Log Analytics workspace	id-sits-test-destroy-swc

Prod destroy identity roles, including lock removal.

Home > Managed Identities > id-sits-prod-destroy-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

Manage view

...

Name ↑

id-sits-prod-destroy-swc

id-sits-prod-destroy-swc | Azure role assignments

Managed Identity

Search

+ Add role assignment (Preview)

Refresh

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
Contributor	rg-sits-prod-network-swc	Resource Group	id-sits-prod-destroy-swc	None
Contributor	rg-sits-prod-app-swc	Resource Group	id-sits-prod-destroy-swc	None
Contributor	rg-sits-prod-kv-swc	Resource Group	id-sits-prod-destroy-swc	None
SITS Lock Remover	rg-sits-prod-kv-swc	Resource Group	id-sits-prod-destroy-swc	None
SITS Lock Remover	rg-sits-prod-app-swc	Resource Group	id-sits-prod-destroy-swc	None
Storage Blob Data Contributor	stsitstfprodswc/default/tfstate-lay...	Microsoft.Storage/storageAccounts/bl...	id-sits-prod-destroy-swc	None
Storage Blob Data Contributor	stsitstfprodswc/default/tfstate-lay...	Microsoft.Storage/storageAccounts/bl...	id-sits-prod-destroy-swc	None
Reader	rg-sits-prod-foundation-swc	Resource Group	id-sits-prod-destroy-swc	None
Network Contributor	NetworkWatcherRG	Resource Group	id-sits-prod-destroy-swc	None

Peter Hudcovic · Azure Platform Lab

68

Dev infra layer 1 identity roles.

 id-sits-dev-infra-l1-swc | Azure role assignments

Managed Identity

Search

◇

«

+ Add role assignment (Preview)

↻ Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 Azure role assignments

 Associated resources (preview)

 Resource visualizer

Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Storage Blob Data Contributor	 stsitsfdevswc/default/tfstate-layer1	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-dev-infra-l1-swc
Contributor	 rg-sits-dev-network-swc	Resource Group	id-sits-dev-infra-l1-swc
Key Vault Contributor	 kv-sits-dev-adm-swc	Key vault	id-sits-dev-infra-l1-swc
Key Vault Contributor	 kv-sits-dev-cert-swc	Key vault	id-sits-dev-infra-l1-swc
Log Analytics Contributor	 log-sits-dev-swc	Log Analytics workspace	id-sits-dev-infra-l1-swc
Reader	 rg-sits-dev-foundation-swc	Resource Group	id-sits-dev-infra-l1-swc
Network Contributor	 NetworkWatcherRG	Resource Group	id-sits-dev-infra-l1-swc
Storage Account Contributor	 stsitsflowdevswc	Storage account	id-sits-dev-infra-l1-swc

Peter Hudcovic · Azure Platform Lab

69

Dev infra layer 2 identity roles.

 **id-sits-dev-infra-l2-swc** | Azure role assignments

Managed Identity

Search

»

+

Add role assignment (Preview)

↺

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Storage Blob Data Contributor	 stsitstfdevswc/default/tfstate-layer2	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-dev-infra-l2-swc
Contributor	 rg-sits-dev-kv-swc	Resource Group	id-sits-dev-infra-l2-swc
Contributor	 rg-sits-dev-app-swc	Resource Group	id-sits-dev-infra-l2-swc
Managed Identity Operator	 id-sits-dev-app-swc	Managed Identity	id-sits-dev-infra-l2-swc
Reader	 rg-sits-dev-foundation-swc	Resource Group	id-sits-dev-infra-l2-swc
Network Contributor	 rg-sits-dev-network-swc	Resource Group	id-sits-dev-infra-l2-swc
Log Analytics Contributor	 log-sits-dev-swc	Log Analytics workspace	id-sits-dev-infra-l2-swc
Managed Identity Operator	 id-sits-dev-agw-swc	Managed Identity	id-sits-dev-infra-l2-swc

Peter Hudcovic · Azure Platform Lab

70

Test infra layer 1 identity roles.

 **id-sits-test-infra-l1-swc** | Azure role assignments

Managed Identity

Search

◊ <<

[+ Add role assignment \(Preview\)](#) [Refresh](#)

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription ^{*}
sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Contributor	 rg-sits-test-network-swc	Resource Group	id-sits-test-infra-l1-swc
Storage Blob Data Contributor	 stsitstftestswc/default/tfstate-layer1	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-test-infra-l1-swc
Log Analytics Contributor	 log-sits-dev-swc	Log Analytics workspace	id-sits-test-infra-l1-swc
Key Vault Contributor	 kv-sits-test-cert-swc	Key vault	id-sits-test-infra-l1-swc
Key Vault Contributor	 kv-sits-test-adm-swc	Key vault	id-sits-test-infra-l1-swc
Reader	 rg-sits-test-foundation-swc	Resource Group	id-sits-test-infra-l1-swc
Network Contributor	 NetworkWatcherRG	Resource Group	id-sits-test-infra-l1-swc
Storage Account Contributor	 stsitsflowtestswc	Storage account	id-sits-test-infra-l1-swc

Test infra layer 2 identity roles.

 **id-sits-test-infra-l2-swc** | Azure role assignments

Managed Identity

Search

»

+

Add role assignment (Preview)

↺

Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

- Federated credentials
- Properties
- Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-nonprod

Role	Resource Name	Resource Type	Assigned To
Contributor	 rg-sits-test-app-swc	Resource Group	id-sits-test-infra-l2-swc
Contributor	 rg-sits-test-kv-swc	Resource Group	id-sits-test-infra-l2-swc
Storage Blob Data Contributor	 stsitstfttestswc/default/tfstate-layer2	Microsoft.Storage/storageAccounts/blobServices/...	id-sits-test-infra-l2-swc
Managed Identity Operator	 id-sits-test-agw-swc	Managed Identity	id-sits-test-infra-l2-swc
Managed Identity Operator	 id-sits-test-app-swc	Managed Identity	id-sits-test-infra-l2-swc
Network Contributor	 rg-sits-test-network-swc	Resource Group	id-sits-test-infra-l2-swc
Log Analytics Contributor	 log-sits-dev-swc	Log Analytics workspace	id-sits-test-infra-l2-swc
Reader	 appi-sits-dev-swc	Application Insights	id-sits-test-infra-l2-swc
Reader	 rg-sits-test-foundation-swc	Resource Group	id-sits-test-infra-l2-swc

Prod infra layer 1 identity roles.

Home > Managed Identities > id-sits-prod-infra-l1-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

⚙️ Manage view

⋮

☐

Name ↑

☐

 id-sits-prod-infra-l1-swc

 id-sits-prod-infra-l1-swc | Azure role assignments

Managed Identity

⋄

⏪

+ Add role assignment (Preview)

🔄 Refresh

Overview

Activity log

Access control (IAM)

Tags

Azure role assignments

Associated resources (preview)

Resource visualizer

Settings

Federated credentials

Properties

Locks

Monitoring

Automation

Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
Contributor	 rg-sits-prod-network-swc	Resource Group	id-sits-prod-infra-l1-swc	None
Storage Blob Data Contributor	 stsitsfprodswc/default/tfstate-lay...	Microsoft.Storage/storageAccounts/bl...	id-sits-prod-infra-l1-swc	None
Key Vault Contributor	 kv-sits-prod-cert-swc	Key vault	id-sits-prod-infra-l1-swc	None
Reader	 rg-sits-prod-foundation-swc	Resource Group	id-sits-prod-infra-l1-swc	None
Storage Account Contributor	 stsitsflowprodswc	Storage account	id-sits-prod-infra-l1-swc	None
Log Analytics Contributor	 log-sits-prod-swc	Log Analytics workspace	id-sits-prod-infra-l1-swc	None
Network Contributor	 NetworkWatcherRG	Resource Group	id-sits-prod-infra-l1-swc	None
Key Vault Contributor	 kv-sits-prod-adm-swc	Key vault	id-sits-prod-infra-l1-swc	None

Peter Hudcovic · Azure Platform Lab

73

Prod infra layer 2 identity roles, including lock creation.

Home > Managed Identities > id-sits-prod-infra-l2-swc

Managed Ident...

Default Directory (peterhudcovicgmail.onmicrosoft.c...

+ Create

⚙️ Manage view

⋮

☐

Name ↑

☐

 id-sits-prod-infra-l2-swc

 id-sits-prod-infra-l2-swc | Azure role assignments

Managed Identity

⋄

⏪

+ Add role assignment (Preview)

🔄 Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Azure role assignments**

 Associated resources (preview)

 Resource visualizer

⌵ Settings

 Federated credentials

 Properties

 Locks

> Monitoring

> Automation

> Help

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *

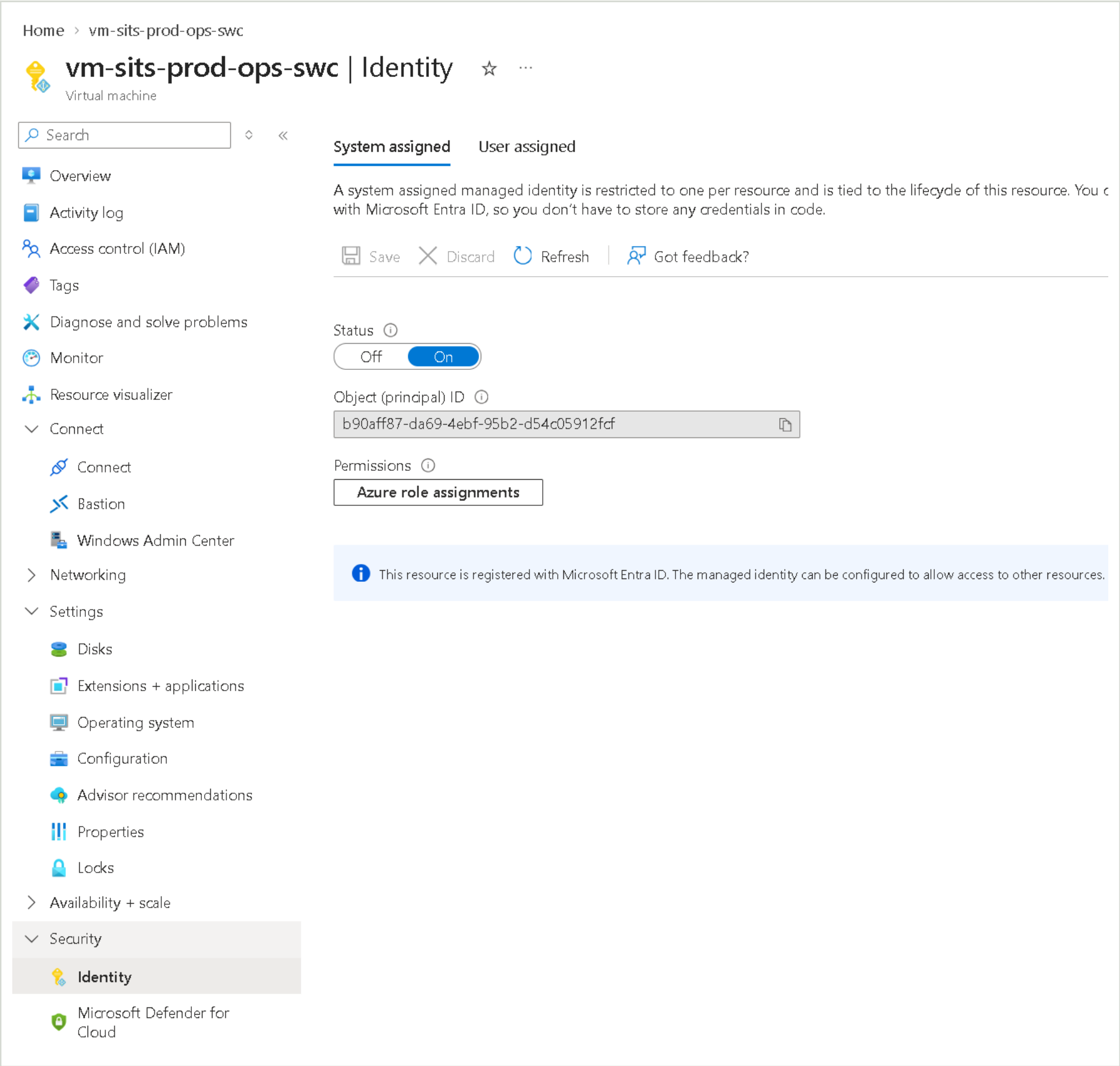
sub-app-prod

Role	Resource Name	Resource Type	Assigned To	Condition
SITS Lock Creator	 rg-sits-prod-app-swc	Resource Group	id-sits-prod-infra-l2-swc	None
Contributor	 rg-sits-prod-kv-swc	Resource Group	id-sits-prod-infra-l2-swc	None
Contributor	 rg-sits-prod-app-swc	Resource Group	id-sits-prod-infra-l2-swc	None
SITS Lock Creator	 rg-sits-prod-kv-swc	Resource Group	id-sits-prod-infra-l2-swc	None
Storage Blob Data Contributor	 stsitstfprodswc/default/tfstate-lay...	Microsoft.Storage/storageAccounts/bl...	id-sits-prod-infra-l2-swc	None
Reader	 rg-sits-prod-foundation-swc	Resource Group	id-sits-prod-infra-l2-swc	None
Network Contributor	 rg-sits-prod-network-swc	Resource Group	id-sits-prod-infra-l2-swc	None
Log Analytics Contributor	 log-sits-prod-swc	Log Analytics workspace	id-sits-prod-infra-l2-swc	None
Managed Identity Operator	 id-sits-prod-agw-swc	Managed Identity	id-sits-prod-infra-l2-swc	None
Managed Identity Operator	 id-sits-prod-app-swc	Managed Identity	id-sits-prod-infra-l2-swc	None

Peter Hudcovic · Azure Platform Lab

74

Prod Ops VM has a system-assigned identity for Entra sign-in.



7 · Networking

Dev VNet 10.10.0.0/16: one subnet per purpose.

Home > vnet-sits-dev-swc

 **vnet-sits-dev-swc** | Subnets

Virtual network

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

- Address space
- Connected devices
- Subnets**
- Bastion
- DDoS protection

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-agw	10.10.4.0/24	-	availability dependent on ...	Microsoft.Network/applic...	nsg-sits-dev-agw-swc	-		
☐	snet-pe	10.10.5.0/24	-	247	-	nsg-sits-dev-pe-swc	-		
☐	snet-app	10.10.6.0/24	-	251	Microsoft.Web/serverFarms	nsg-sits-dev-app-swc	-		
☐	snet-agents-in...	10.10.2.0/24	-	251	Microsoft.DevOpsInfrastr...	nsg-sits-dev-agents-infra-swc	-		
☐	snet-agents-app	10.10.3.0/24	-	251	Microsoft.DevOpsInfrastr...	nsg-sits-dev-agents-app-swc	-		
☐	snet-ops	10.10.1.0/24	-	250	-	nsg-sits-dev-ops-swc	-		

Peter Hudcovic · Azure Platform Lab

77

Test VNet 10.20.0.0/16: one subnet per purpose.

Home > vnet-sits-test-swc

vnet-sits-test-swc | Subnets

Virtual network

Search

+ Subnet Refresh | Manage users Delete Export to CSV

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-agw	10.20.4.0/24	-	availability dependent on ...	Microsoft.Network/applic...	nsg-sits-test-agw-swc	-		
☐	snet-ops	10.20.1.0/24	-	250	-	nsg-sits-test-ops-swc	-		
☐	snet-pe	10.20.5.0/24	-	247	-	nsg-sits-test-pe-swc	-		
☐	snet-app	10.20.6.0/24	-	251	Microsoft.Web/serverFarms	nsg-sits-test-app-swc	-		
☐	snet-agents-app	10.20.3.0/24	-	251	Microsoft.DevOpsInfrastr...	nsg-sits-test-agents-app-swc	-		
☐	snet-agents-infra	10.20.2.0/24	-	251	Microsoft.DevOpsInfrastr...	nsg-sits-test-agents-infra-swc	-		

Prod VNet 10.30.0.0/16: one subnet per purpose, plus AzureBastionSubnet.

Home > vnet-sits-prod-swc

 **vnet-sits-prod-swc** | Subnets

Virtual network

Search

Subnet Refresh Manage users Delete Export to CSV

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-ops	10.30.1.0/24	-	250	-	nsg-sits-pr...	-		
☐	snet-app	10.30.6.0/24	-	251	Microsoft.Web...	nsg-sits-pr...	-		
☐	snet-agents-app	10.30.3.0/24	-	251	Microsoft.Dev...	nsg-sits-pr...	-		
☐	snet-pe	10.30.5.0/24	-	247	-	nsg-sits-pr...	-		
☐	snet-agw	10.30.4.0/24	-	availability dependent on ...	Microsoft.Net...	nsg-sits-pr...	-		
☐	snet-agents-infra	10.30.2.0/24	-	251	Microsoft.Dev...	nsg-sits-pr...	-		
☐	AzureBastionSubnet	10.30.0.0/26	-	57	-	nsg-sits-pr...	-		

Dev connected devices: Ops VM, gateway 10.10.4.10 and private endpoints.

 **vnet-sits-dev-swc** | Connected devices

Virtual network

Refresh

Export To CSV

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Type : all

Subnet : all

	Device ↕	Type ↕	IP Address ↕	Subnet ↕
	 appgwLB	Load balancer	-	snet-agw
	 agw-sits-dev-swc	Application Gateway	10.10.4.10	snet-agw
	 agw-sits-dev-swc	Application Gateway	-	snet-agw
	 nic-pe-app-sits-dev-swc	Network interface	10.10.5.7	snet-pe
	 nic-pe-kv-sits-dev-app-swc	Network interface	10.10.5.6	snet-pe
	 nic-pe-kv-sits-dev-adm-swc	Network interface	10.10.5.5	snet-pe
	 nic-pe-kv-sits-dev-cert-swc	Network interface	10.10.5.4	snet-pe
	 nic-vm-sits-dev-ops-swc	Network interface	10.10.1.4	snet-ops

Test connected devices: Ops VM, gateway 10.20.4.10 and private endpoints.

Microsoft Azure

Search resources, services, and docs (G+)

Home > vnet-sits-test-swc

 **vnet-sits-test-swc** | Connected devices

Virtual network

Search

Refresh

Export To CSV

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for

Search connected devices

Type : all

Subnet : all

	Device ↕	Type ↕	IP Address ↕	Subnet ↕
	 appgwLB	Load balancer	-	snet-agw
	 agw-sits-test-swc	Application Gateway	10.20.4.10	snet-agw
	 agw-sits-test-swc	Application Gateway	-	snet-agw
	 nic-vm-sits-test-ops-swc	Network interface	10.20.1.4	snet-ops
	 nic-pe-app-sits-test-swc	Network interface	10.20.5.7	snet-pe
	 nic-pe-kv-sits-test-app-swc	Network interface	10.20.5.6	snet-pe
	 nic-pe-kv-sits-test-adm-swc	Network interface	10.20.5.4	snet-pe
	 nic-pe-kv-sits-test-cert-swc	Network interface	10.20.5.5	snet-pe

Prod connected devices: Ops VM, gateway 10.30.4.10 and private endpoints.

Home > vnet-sits-prod-swc

 **vnet-sits-prod-swc** | Connected devices

Virtual network

Refresh

Export To CSV

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Type : all

Subnet : all

	Device ↕	Type ↕	IP Address ↕	Subnet ↕
☐	 nic-vm-sits-prod-ops-swc	Network interface	10.30.1.4	snet-ops
☐	 nic-pe-app-sits-prod-swc	Network interface	10.30.5.7	snet-pe
☐	 nic-pe-kv-sits-prod-app-swc	Network interface	10.30.5.6	snet-pe
☐	 nic-pe-kv-sits-prod-adm-swc	Network interface	10.30.5.5	snet-pe
☐	 nic-pe-kv-sits-prod-cert-swc	Network interface	10.30.5.4	snet-pe
☐	 appgwLB	Load balancer	-	snet-agw
☐	 agw-sits-prod-swc	Application Gateway	10.30.4.10	snet-agw
☐	 agw-sits-prod-swc	Application Gateway	-	snet-agw
☐	 bas-sits-prod-swc	Bastion	-	AzureBastionSubnet

Ops NSG: RDP only from Bastion, then deny-all at priority 4000.

Home

 **nsg-sits-prod-ops-swc**

Network security group

☆ ☆ ...

How do I create an alert to track firewall metric failures

✕

Search

◇ <<

→ Move ▾ 🗑️ Delete ↻ Refresh 🗨️ Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

> Settings

> Monitoring

> Automation

> Help

^ Essentials

JSON View

Resource group (move) : rg-sits-prod-network-swc

Location : Sweden Central

Subscription (move) : sub-app-prod

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags (edit) : costCenter : sits-lab environment : prod owner : peter

Custom security rules : 2 inbound, 1 outbound

Associated with : 1 subnets, 0 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓	
Inbound Security Rules							
100	 allow-rdp-from-bastion	3389	Tcp	10.30.0.0/26	 asg-sits-prod-ops-vm-s...	 Allow	
4000	 deny-all-inbound	Any	Any	Any	Any	 Deny	
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow	
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any	 Allow	
65500	DenyAllInBound	Any	Any	Any	Any	 Deny	
Outbound Security Rules							
100	 deny-ops-vm-to-azure-r...	Any	Any	 asg-sits-prod-ops-vm-s...	AzureResourceManager	 Deny	
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow	
65001	AllowInternetOutBound	Any	Any	Any	Internet	 Allow	
65500	DenyAllOutBound	Any	Any	Any	Any	 Deny	

Private endpoint NSG: only HTTPS to named targets, then deny-all.

 **nsg-sits-prod-pe-swc**

Network security group

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : rg-sits-prod-network-swc

Location : Sweden Central

Subscription (move) : sub-app-prod

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags (edit) : costCenter : sits-lab environment : prod owner : peter

Custom security rules : 8 inbound, 0 outbound

Associated with : 1 subnets, 0 network interfaces

JSON View

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓
Inbound Security Rules						
100	 allow-webapp-from-agw	443	Tcp	10.30.4.0/24	 asg-sits-prod-pe-webap...	 Allow 
110	 allow-webapp-from-age...	443	Tcp	10.30.3.0/24	 asg-sits-prod-pe-webap...	 Allow 
120	 allow-kv-app-from-app	443	Tcp	10.30.6.0/24	 asg-sits-prod-pe-kv-app...	 Allow 
130	 allow-kv-app-from-agen...	443	Tcp	10.30.2.0/24	 asg-sits-prod-pe-kv-app...	 Allow 
140	 allow-kv-cert-from-agw	443	Tcp	10.30.4.0/24	 asg-sits-prod-pe-kv-cert...	 Allow 
150	 allow-kv-cert-from-agen...	443	Tcp	10.30.2.0/24	 asg-sits-prod-pe-kv-cert...	 Allow 
160	 allow-kv-admin-from-ag...	443	Tcp	10.30.2.0/24	 asg-sits-prod-pe-kv-ad...	 Allow 
4000	 deny-all-inbound	Any	Any	Any	Any	 Deny 
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow 
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any	 Allow 
65500	DenyAllInBound	Any	Any	Any	Any	 Deny 
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	 Allow 
65001	AllowInternetOutBound	Any	Any	Any	Internet	 Allow 
65500	DenyAllOutBound	Any	Any	Any	Any	 Deny 

Private endpoint subnet with network policies enabled.

Home > vnet-sits-prod-swc

vnet-sits-prod-swc

Subnets

Virtual network

Search

Subnet Refresh Manage users Delete Export to CSV

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS

Peerings

Service endpoints

Private endpoints

Properties

Locks

Monitoring

Automation

Help

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you create a subnet, you can specify the address range, size, and other properties.

Search subnets

	Name ↑	IPv4	IPv6	Availability
☐	snet-ops	10.30.1.0/24	-	250
☐	snet-app	10.30.6.0/24	-	251
☐	snet-agents-app	10.30.3.0/24	-	251
☒	snet-pe	10.30.5.0/24	-	247
☐	snet-agw	10.30.4.0/24	-	available
☐	snet-agents-infra	10.30.2.0/24	-	251
☐	AzureBastionSubnet	10.30.0.0/26	-	57

Showing 7 subnets

Add or remove favorites by pressing Ctrl+Shift+F

Edit subnet

Include an IPv4 address space

☒ Address prefix '10.30.5.0/24' cannot be updated because the subnet is in use.

IPv4 address space *

10.30.0.0/16

Starting address *

10.30.5.0

Size

/24

Subnet address range

10.30.5.0 - 10.30.5.255

IPv6

☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Enable private subnet (no default outbound access)

☒

After March 31, 2026, private subnet will be the default selection for new virtual networks. [Learn more](#)

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security group. [Learn more](#)

NAT gateway

None

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

Network security group

nsg-sits-prod-pe-swc

Route table

None

Service Endpoints

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Network identifier subscription

sub-app-prod

Network identifier

None

Services

Enable network identifier

Remove service endpoint

Select a service endpoint

Subnet Delegation

Delegate subnet to a service

None

Network Policy for Private Endpoints

The network policy affects the types of network policies that control traffic going to the private endpoints in this subnet. [Learn more](#)

Private endpoint network policy

Route tables, Network security groups

Save

Cancel

[Give feedback](#)

NAT Gateway serves only the ops, agent and app subnets.

Home > ng-sits-prod-swc

<>

ng-sits-prod-swc | Networking

☆

...

NAT gateway

Search

◇

«

Save

✕ Discard

🗑 Disassociate

🔄 Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Outbound IP

Networking

Configuration

Properties

Locks

Monitoring

Automation

Help

To use the NAT gateway, at least one subnet must be selected. You can add and remove subnets after creating the NAT gateway.

Virtual network ⓘ

vnet-sits-prod-swc

Create new

Subnets that have any of the following resources are not shown because they are not compatible:

- A load balancer with a Basic SKU
- A public IP address with a Basic SKU
- An existing NAT gateway
- A virtual network gateway

☐ Subnet name	Subnet address range
☒ snet-app	10.30.6.0/24
☒ snet-agents-app	10.30.3.0/24
☒ snet-agents-infra	10.30.2.0/24
☒ snet-ops	10.30.1.0/24
☐ snet-pe	10.30.5.0/24
☐ snet-agw	10.30.4.0/24
☐ AzureBastionSubnet	10.30.0.0/26

Manage subnets >

Prod NAT Gateway overview.

Home

 **ng-sits-prod-swc**
NAT gateway

  ... 

[Run health check for my NAT Gateway](#) [How to improve resiliency of my nat gateway](#) [Inspect connectivity for my NAT Gateway](#)

 Overview

 Activity log

 Access control (IAM)

 Tags

 Diagnose and solve problems

 Resource visualizer

Settings

 Outbound IP

 Networking

 Configuration

 Properties

 Locks

> Monitoring

> Automation

> Help

Delete

Refresh

^ Essentials

Resource group [\(move\)](#) : [rg-sits-prod-network-swc](#)

Location : Sweden Central (Zone 1)

Subscription [\(move\)](#) : [sub-app-prod](#)

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags [\(edit\)](#) :

costCenter : sits-lab

environment : prod

owner : peter

Virtual network : [vnet-sits-prod-swc](#)

Subnets : [4](#)

SKU : Standard

Public IP addresses : [1](#)

Public IP prefixes : 0

 **Configure outbound IP addresses**
Configure which public IP addresses and public IP prefixes to use for outbound connectivity.

 **Configure networking**
Configure which subnets of a virtual network should use this NAT gateway.

Peter Hudcovic · Azure Platform Lab

87

NAT Gateway outbound public IP.

Home > ng-sits-prod-swc

 **ng-sits-prod-swc** | Outbound IP

NAT gateway

Search

Edit

Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Outbound IP

Networking

Configuration

Properties

Locks

Monitoring

Automation

Help

View and configure which public IP addresses and public IP prefixes will be used for outbound connectivity. At least one is required for outbound connectivity.

Public IP addresses

Name ↑↓	IP address ↑↓	IP version ↑↓	Remove
pip-sits-prod-nat-swc	172.160.250.161	IPv4	

Public IP prefixes

Name ↑↓	IP prefix ↑↓	IP version ↑↓	Remove
No items found			

NAT public IP resource.

Home > ng-sits-prod-swc | Outbound IP

 **pip-sits-prod-nat-swc**
Public IP address

☆

...



Can you check if my public Ip is highly available

How to ensure public Ip fault tolerance

Search

⬵

⬅

 Associate

 Dissociate

 Delete

 Move

⌵

 Refresh

 Open in mobile

 Give feedback

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move)

Location (move)

Subscription (move)

Subscription ID

Tags (edit)

See more

Get Started

Properties

Tutorials

SKU

Tier

IP address

DNS name

Associated to

Virtual machine

: Standard

: Regional

: 172.160.250.161

: -

: ng-sits-prod-swc

: -

costCenter : sits-lab

environment : prod

owner : peter

Peter Hudcovic · Azure Platform Lab

89

8 - Private DNS and isolation proof

Nine private DNS zones, each linked to exactly one VNet (1 / 1000).

Home > Network foundation

Network foundation | Private DNS zones

Preview

Filter Private DNS zones by resource group and location.

Sort Private DNS zones by creation date.

Identify non-compliant Private DNS zones in my environment.

Search

CreateManage viewRefreshExport to CSVOpen queryAssign tagsDeleteAdd to service group

Group by none

Overview

Virtual network

Private Link

DNS

DNS overview

DNS private resolvers

Public DNS zones

Private DNS zones

Traffic managers

DNS security policies

Related services

Monitoring and management

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

☐	Name ↑		Number of record sets	Max number of record ...	Number of virtual net...	Number of virtual net...	Resource Group	Subscription
☐	dev.sits.internal	...	2	25,000	1 / 1000	0 / 100	rg-sits-dev-network-swc	sub-app-nonprod
☐	privatelink.azurewebsites.net	...	3	25,000	1 / 1000	0 / 100	rg-sits-prod-network-s...	sub-app-prod
☐	privatelink.azurewebsites.net	...	3	25,000	1 / 1000	0 / 100	rg-sits-dev-network-swc	sub-app-nonprod
☐	privatelink.azurewebsites.net	...	3	25,000	1 / 1000	0 / 100	rg-sits-test-network-swc	sub-app-nonprod
☐	privatelink.vaultcore.azure.net	...	4	25,000	1 / 1000	0 / 100	rg-sits-prod-network-s...	sub-app-prod
☐	privatelink.vaultcore.azure.net	...	4	25,000	1 / 1000	0 / 100	rg-sits-dev-network-swc	sub-app-nonprod
☐	privatelink.vaultcore.azure.net	...	4	25,000	1 / 1000	0 / 100	rg-sits-test-network-swc	sub-app-nonprod
☐	prod.sits.internal	...	2	25,000	1 / 1000	0 / 100	rg-sits-prod-network-s...	sub-app-prod
☐	test.sits.internal	...	2	25,000	1 / 1000	0 / 100	rg-sits-test-network-swc	sub-app-nonprod

prod.sits.internal: app points to the Prod gateway 10.30.4.10.

Home > prod.sits.internal

 **prod.sits.internal | Recordsets** ☆ ...

Private DNS zone

Search

⌵ ⌵

+ Add

🔄 Refresh

🗑 Delete

🗨 Give feedback

📄 Overview

📅 Activity log

👤 Access control (IAM)

🏷 Tags

🔧 Diagnose and solve problems

📊 Resource visualizer

> Settings

▼ DNS Management

 **Recordsets**

🔗 Virtual Network Links

> Monitoring

> Automation

> Help

Search

A record set is a collection of records in a zone that have the same name and are the same type. Record Sets will be automatically fetched in batches of 100 as you scroll through the existing record sets. [Learn more](#)

Search

Fetches 2 record set(s).

0 record sets selected

Name	Type	TTL	Value	Auto registered
@	SOA	3600	Email: azureprivatedns-host.microsoft.com Host: azureprivatedns.net Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 10 Serial number: 1	False
app	A	300	10.30.4.10	False

Web App private DNS zone is linked only to the Prod VNet.

Home > pe-app-sits-prod-swc | DNS configuration > privatelink.azurewebsites.net

 **privatelink.azurewebsites.net** | Virtual Network Links

Private DNS zone

+ Add

Refresh

Delete

Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

DNS Management

Recordsets

Virtual Network Links

Monitoring

Automation

Help

Fetches 1 virtual Network link(s).

0 Virtual Network links selected

Link Name	Link Status	Virtual Network	Auto-Registration	Fallback to Internet
link-webapp-vnet-sits-prod-swc	Completed	vnet-sits-prod-swc	Disabled	Disabled

Key Vault private DNS zone is linked only to the Prod VNet.

Home > pe-kv-sits-prod-app-swc | DNS configuration > privatelink.vaultcore.azure.net

 **privatelink.vaultcore.azure.net** | Virtual Network Links

Private DNS zone

Search

⬅

+ Add

🔄 Refresh

🗑 Delete

🗨 Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

DNS Management

Recordsets

Virtual Network Links

Monitoring

Automation

Help

Search virtual network links

Fetches 1 virtual Network link(s).

0 Virtual Network links selected

Link Name	Link Status	Virtual Network	Auto-Registration	Fallback to Internet
link-keyvault-vnet-sits-prod-swc	Completed	vnet-sits-prod-swc	Disabled	Disabled

Peter Hudcovic · Azure Platform Lab

94

Web App private endpoint DNS configuration.

Home > pe-app-sits-prod-swc

pe-app-sits-prod-swc | DNS configuration

Private endpoint

Search

+ Add configuration

Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Application security groups

DNS configuration

Properties

Locks

Monitoring

Automation

Help

Private DNS integration

To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint using a private DNS zone. You can also utilize your own DNS servers. [Learn more](#)

Customer Visible FQDNs

DNS records visible to the customer

Network Interface	IP addresses	FQDN
▼ nic-pe-app-sits-prod-swc		
▼	10.30.5.7	
		app-sits-prod-swc.azurewebsites.net
		app-sits-prod-swc.scm.azurewebsites.net

Configuration name	FQDN	IP address	Subscription	Private DNS zone	DNS zone group
▼ privatelink.azurewebsites.net			sub-app-prod	privatelink.azurewebsites.net	default
▼	app-sits-prod-swc.privatelink.azurewebsites.net			-	-
		10.30.5.7		-	-
▼	app-sits-prod-swc.scm.privatelink.azurewebsites.net			-	-
		10.30.5.7		-	-

App Key Vault private endpoint DNS configuration.

Home > pe-kv-sits-prod-app-swc

DNS

pe-kv-sits-prod-app-swc | DNS configuration

Private endpoint

☆

...

Search

⋄

«

+ Add configuration

🔄 Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Application security groups

DNS configuration

Properties

Locks

Monitoring

Automation

Help

Private DNS integration

To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint using a private DNS zone. You can also utilize your own DNS servers. [Learn more](#)

Customer Visible FQDNs

DNS records visible to the customer

Network Interface	IP addresses	FQDN
▼ nic-pe-kv-sits-prod-app-swc		
▼	10.30.5.6	
		kv-sits-prod-app-swc.vault.azure.net

Configuration name	FQDN	IP address	Subscription	Private DNS zone	DNS zone group	
▼ privatelink.vaultcore.azure....			sub-app-prod	privatelink.vaultcore.azure.net	default	
▼	kv-sits-prod-app-swc.privatedlink.vaultcore.azure.net			-	-	
		10.30.5.6		-	-	

Peter Hudcovic · Azure Platform Lab

96

From the Dev VM the Prod name does not exist; the Dev name resolves to 10.10.4.10.

```
PS C:\Users\opsadmin> Resolve-DnsName app.prod.sits.internal
Resolve-DnsName : app.prod.sits.internal : DNS name does not exist
At line:1 char:4
+ Resolve-DnsName app.prod.sits.internal
+ ~~~~~
+ CategoryInfo          : ResourceUnavailable: (app.prod.sits.internal:String) [Resolve-DnsName], Win32Exception
+ FullyQualifiedErrorId : DNS_ERROR_RCODE_NAME_ERROR,Microsoft.DnsClient.Commands.ResolveDnsName

PS C:\Users\opsadmin> Resolve-DnsName app.dev.sits.internal

Name                                     Type  TTL  Section  IPAddress
----
app.dev.sits.internal                   A     300  Answer   10.10.4.10

PS C:\Users\opsadmin> _
```

From the Prod VM: DNS resolves to 10.30.4.10 and port 443 is reachable.

```
PS C:\Users\opsadmin> Resolve-DnsName app.prod.sits.internal -Type A

Name                                     Type  TTL  Section  IPAddress
----
app.prod.sits.internal                  A      300  Answer   10.30.4.10

PS C:\Users\opsadmin> Test-NetConnection app.prod.sits.internal -Port 443

ComputerName      : app.prod.sits.internal
RemoteAddress     : 10.30.4.10
RemotePort        : 443
InterfaceAlias    : Ethernet
SourceAddress     : 10.30.1.4
TcpTestSucceeded  : True

PS C:\Users\opsadmin> Resolve-DnsName app-sits-prod-swc.azurewebsites.net -Type A

Name                                     Type  TTL  Section  NameHost
----
app-sits-prod-swc.azurewebsite CNAME  60   Answer   app-sits-prod-swc.privatelink.azurewebsites.net
s.net

Name      : app-sits-prod-swc.privatelink.azurewebsites.net
QueryType : A
TTL       : 10
Section   : Answer
IP4Address : 10.30.5.7

PS C:\Users\opsadmin> Resolve-DnsName kv-sits-prod-app-swc.vault.azure.net -Type A

Name                                     Type  TTL  Section  NameHost
----
kv-sits-prod-app-swc.vault.azu CNAME  60   Answer   kv-sits-prod-app-swc.privatelink.vaultcore.azure.net
re.net

Name      : kv-sits-prod-app-swc.privatelink.vaultcore.azure.net
QueryType : A
TTL       : 10
Section   : Answer
IP4Address : 10.30.5.6
```

Prod admin session: DNS and gateway connectivity confirmed.

```
Name                                Type      TTL      Section    IPAddress
-----
app.prod.sits.internal              A          300      Answer     10.30.4.10

ComputerName                        : app.prod.sits.internal
RemoteAddress                       : 10.30.4.10
ResolvedAddresses                   : <10.30.4.10>
PingSucceeded                       : False
PingReplyDetails                    :
TcpClientSocket                     :
TcpTestSucceeded                    : True
RemotePort                          : 443
TraceRoute                          :
Detailed                            : False
InterfaceAlias                      : Ethernet
InterfaceIndex                      : 6
InterfaceDescription                : Microsoft Hyper-V Network Adapter
NetAdapter                          : MSFT_NetAdapter <CreationClassName = "MSFT_NetAdapter", DeviceID = "<CDCC59CD-8A52-4E59-A8A9-EDB73BC7F81A>", SystemCreationClassName = "CIM_NetworkPort", SystemName =
NetRoute                            : MSFT_NetRoute <InstanceID = ":8:8:8:9:55055;:8?:8;8;55;">
SourceAddress                       : 10.30.1.4
NameResolutionSucceeded             : True
BasicNameResolution                 : <Microsoft.DnsClient.Commands.DnsRecord_A>
LLMNRNetbiosRecords                 : <>
DNSOnlyRecords                      : <Microsoft.DnsClient.Commands.DnsRecord_A>
AllNameResolutionResults             : Microsoft.DnsClient.Commands.DnsRecord_A
IsAdmin                             : False
NetworkIsolationContext              : Internet
MatchingIPsecRules                  :

Name      : app-sits-prod-swc.azurewebsites.net
QueryType : CNAME
TTL       : 60
Section   : Answer
NameHost  : app-sits-prod-swc.privatelink.azurewebsites.net

app-sits-prod-swc.privatelink.azurewebsites.ne A      10      Answer     10.30.5.7
t

Name      : kv-sits-prod-app-swc.vault.azure.net
QueryType : CNAME
TTL       : 34
Section   : Answer
NameHost  : kv-sits-prod-app-swc.privatelink.vaultcore.azure.net

kv-sits-prod-app-swc.privatelink.vaultcore.azu A      10      Answer     10.30.5.6
re.net

PS C:\Users\ProdAdmin> A_
```

9 - Application Gateway, WAF and TLS

Prod Application Gateway WAF_v2 with a private frontend only.

Home

 **agw-sits-prod-swc**
Application gateway

 ... 

Run diagnostic for my Application Gateway

Analyze traffic to this gateway

Check health of application gateway

Search

⌵

⏪

 Delete

 Refresh

 Feedback

 Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

^ Essentials

Resource group [\(move\)](#) : [rg-sits-prod-app-swc](#)

Location : Sweden Central

Subscription [\(move\)](#) : [sub-app-prod](#)

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags [\(edit\)](#) :

costCenter : sits-lab

environment : prod

Show more (1)

Virtual network/subnet : [vnet-sits-prod-swc / snet-agw](#)

Frontend public IP address : -

Frontend private IP address : 10.30.4.10

Tier : WAF V2

Availability zone : -

Recommendations

Metrics



HTTPS listener on 443 for app.prod.sits.internal.

Home > agw-sits-prod-swc

 agw-sits-prod-swc | Listeners

Application gateway

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Web application firewall

Backend pools

Backend settings

Frontend IP configurations

Private link

JWT validation configurations (Preview)

SSL settings

Listeners

Rules

Rewrites

Health probes

Properties

Listeners

Listener TLS certificates

+ Add listener

Refresh

Feedback

Troubleshoot

Application Gateway provides native support for WebSocket across all gateway sizes. There is no additional configuration required to enable or disable WebSocket support. If a WebSocket traffic is received on the Application Gateway, it is automatically directed to the WebSocket enabled backend server using the appropriate backend pool as specified in application gateway rules. [Learn more about listeners and WebSocket support.](#)

Search listeners

Name	Port	Protocol	Frontend IP	Associated rule	Host name
https-prod	443	HTTPS	Private IPv4	app-prod	> app.prod.sits.internal

SSL Policy

The SSL policy defines the SSL protocol version and available ciphers. Choose from one of the predefined policies or create a custom security policy to match your organizational security requirements. These policies apply to all HTTPS listeners unless they are overridden by listener specific SSL Policy under SSL settings. [Learn more about SSL policy.](#)

Selected SSL Policy: Predefined

Policy name
AppGwSslPolicy20220101

Min protocol version
TLSv1_2

Cipher suites
TLS_AES_256_GCM_SHA384
TLS_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384

Peter Hudcovic · Azure Platform Lab

102

One routing rule for the Prod application.

Home > agw-sits-prod-swc

 **agw-sits-prod-swc** | Rules

Application gateway

 Routing rule

 Backend health

 Feedback

 Overview

 Activity log

 Access control (IAM)

 Tags

 Diagnose and solve problems

 Resource visualizer

Settings

 Configuration

 Web application firewall

 Backend pools

 Backend settings

 Frontend IP configurations

 Private link

 JWT validation configurations (Preview)

 SSL settings

 Listeners

 **Rules**

 Search rules

Name	↑↓	Type	↑↓	Listener	↑↓	Priority	↑↓
app-prod		Basic		https-prod		100	...

Peter Hudcovic · Azure Platform Lab

103

Routing rule uses the HTTPS listener.

[Home](#) > [agw-sits-prod-swc](#) | Rules

app-prod

...

1 Listener

2 Routing

3 Review

Configure a routing rule to send traffic from a listener to one or more backend targets.

Rule name *

app-prod

Rule Priority * ⓘ

100

Listener

A listener accepts traffic on a specified IP address, port, and protocol. Matching requests are routed according to the associated rule.

Listener *

https-prod

▼

[Create new](#)

Routing rule sends traffic to the Web App backend.

Home > agw-sits-prod-swc | Rules

app-prod ...

1 Listener

2 Routing

3 Review

A routing rule defines how requests are matched, routed, and load-balanced across backend resources. [Learn more](#)

Rule type *

☒ Basic

Routes traffic without evaluating request attributes or matching conditions.

☐ Advanced (Preview)

Route traffic based on Header conditions. More condition types coming soon.

☐ Path-based

Route traffic based on Path conditions.

✓ Backend targets

Select the backend target that must serve the traffic.

BASIC RULE

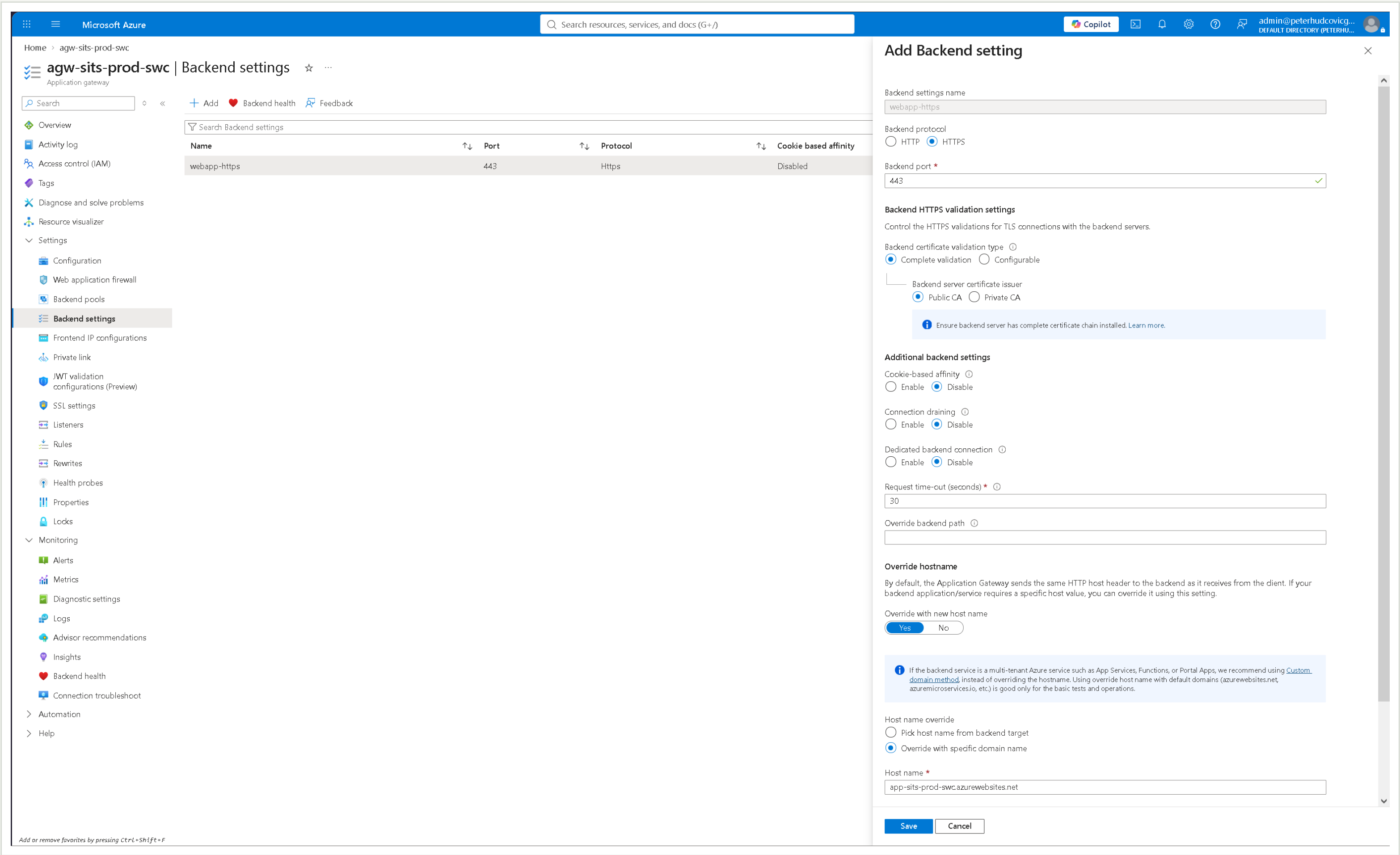
Route traffic to **backend pool**

Target	Backend setting
webapp	webapp-https ...

Peter Hudcovic · Azure Platform Lab

105

Backend uses HTTPS 443 with full certificate validation.



Custom health probe checks /health over HTTPS.

Home > agw-sits-prod-swc

 **agw-sits-prod-swc | Health**
Application gateway

<< + Add

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Web application firewall

Backend pools

Backend settings

Frontend IP configurations

Private link

JWT validation configurations (Preview)

SSL settings

Listeners

Rules

Rewrites

Health probes

webapp-health

agw-sits-prod-swc

Name

webapp-health

Protocol *

HTTP

HTTPS

TCP

TLS

Pick host name from backend settings

Yes

No

Host *

app-sits-prod-swc.azurewebsites.net

Pick port from backend settings

Yes

No

Path *

/health

Interval (seconds) *

30

Timeout (seconds) *

30

Unhealthy threshold *

3

Use probe matching conditions

Yes

No

HTTP response status code match *

200-399

HTTP response body match

Backend settings

webapp-https

Peter Hudcovic · Azure Platform Lab

107

Backend health: Healthy.

Home > agw-sits-prod-swc

 agw-sits-prod-swc | Backend health

☆ ...

Application gateway

Search

◇ <<

 Refresh

 Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Alerts

Metrics

Diagnostic settings

Logs

Advisor recommendations

Insights

Backend health

Connection troubleshoot

Automation

Help

Backend health

By default, Azure Application Gateway probes backend servers to check their health and whether they're ready to serve requests. You can also create custom [Health Probes](#) to mention a specific hostname and path to be probed or a response code to be accepted as Healthy.

 The Backend health report is updated based on the respective probe's refresh interval and doesn't depend on the page refresh.

All

 1 out of 1

Healthy

 1 out of 1

Search backend health

Server (backend pool)	↑↓ Status	↑↓ Port (Backend setting)	↑↓ Protocol	↑↓ Details	Action
app-sits-prod-swc.azurewebsites.net (webapp)	 Healthy	443 (webapp-https)	Https	Success. Received 200 status code	

Peter Hudcovic · Azure Platform Lab

108

Listener certificate comes from Key Vault.

Home > agw-sits-prod-swc

 agw-sits-prod-swc | Listeners

Application gateway

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Web application firewall

Backend pools

Listeners

Listener TLS certificates

+ Add certificate

Refresh

Bulk update

Feedback

This section allows you to view and manage all TLS certificates for listeners. An TLS certificate must be associated with a listener to configure TLS/SSL termination. You can create new certificate objects, re-upload a certificate for renewals or perform other management operations. [Learn more.](#)

Search Listener Certificate

Certificate name	Type	Key vault ID	Status (Expiry)	Common Name	Associated listener
☐ cert-app-prod	Key Vault	https://kv-sits-prod-cert-swc.vault.azure.net/secrets/cert-app-prod	Active (Oct 5, 2027)	app.prod.sits.internal	1

Certificate cert-app-prod is active in the certificate vault.

Listeners

Listeners

Listener TLS certificates

+ Add certificate

Refresh

Bulk update

Feedback

This section allows you to view and manage all TLS certificates for listeners. An TLS certificate must be associated with a listener to configure TLS/SSL termination. You can create new certificate objects, re-upload

Search Listener Certificate

Certificate name	Type	Key vault ID	Status (Expiry)	Common
cert-app-prod	Key Vault	https://kv-sits-prod-cert-swc.vault.azure.net/secrets/cert-app-prod	Active (Oct 5, 2027)	app.prod

agw-sits-prod-swc/cert-app-prod

agw-sits-prod-swc

Edit

Delete

Refresh

Certificate Information

Listener certificate name

cert-app-prod

Type

Key Vault

Copy to clipboard

Key vault ID

https://kv-sits-prod-cert-swc.vault.azure.ne...

Managed identity

/subscriptions/61845030-2daa-4797-a97a...

Status

Active

Expiry

Oct 5, 2027

SHA-1 Thumbprint

0D854E3D562A5CD4B075204CCCCDA2C1...

Subject Common Name

app.prod.sits.internal

Issuer Common Name

app.prod.sits.internal

Subject Alternative Name

app.prod.sits.internal

Associated listener

agw-sits-prod-swc/https-prod

WAF policy enabled in Detection mode.

Home > agw-sits-prod-swc | Web application firewall

**waf-sits-prod-swc**

Application Gateway WAF policy





How do I troubleshoot issues with this Application Gateway WAF policy?

Summarize the configurations of this Application Gateway WAF policy.

List the associated Application Gateways for this policy.

Search

Search

Disable

Switch to prevention mode

Delete

Refresh

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group [\(move\)](#) : rg-sits-prod-app-swc

Location : Sweden Central

Subscription [\(move\)](#) : sub-app-prod

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags [\(edit\)](#) : costCenter : sits-labenvironment : prodowner : peter

Policy state :  Enabled

Policy mode : Detection

Associations : 1

**Policy settings**

Configure multiple settings that apply to all rules within the policy.

**Managed rules**

Configure Azure managed rule sets that protect your web application from common threats.

**Associations**

Associate this WAF policy to one or more application gateways for protection.

**Custom rules**

Author custom rules to protect your web application from specifically identified threats.

**Log scrubbing rules**

Add log scrubbing rules to scrub sensitive data in the logs within this policy.

Peter Hudcovic · Azure Platform Lab

111

WAF policy is attached to the Prod gateway.

 **waf-sits-prod-swc** | Associations  

Application Gateway WAF policy

 Search

 Overview

 Activity log

 Access control (IAM)

 Tags

 Resource visualizer

 Settings

 Policy settings

 Managed rules

 Custom rules

 **Associations**

Associate this WAF policy with a specific application gateway, listener, route path, or application gateway for containers. A WAF policy can be associated with multiple application gateways. You can add, remove, and replace existing WAF policy associations with the selected application gateway. [Learn more about associations](#)

 Deselect application gateway(s). To disassociate the selected application gateway, associate the gateway to a different WAF policy.

 Add association 

 Remove association

 Refresh

 Filter for any field

Association type : **all** 

Application Gateway : **all** 

Resource Group : **all** 

 Association type	Application Gateway	Resource Group
 Application Gateway	agw-sits-prod-swc	rg-sits-prod-app-swc

WAF uses Microsoft Default Rule Set 2.1.

Home > agw-sits-prod-swc | Web application firewall > waf-sits-prod-swc

📄

waf-sits-prod-swc | Managed rules

☆

⋮

Application Gateway WAF policy

🔍 Search

⬢ ⬅

🏠 Overview

📋 Activity log

👤 Access control (IAM)

🏷️ Tags

🔗 Resource visualizer

⌵ Settings

⚙️ Policy settings

📄

Managed rules

📄

Custom rules

🔗

Associations

📄

Sensitive data

📋

Properties

🔒

Locks

🔍

Monitoring

🔍

Automation

🔍

Help

Managed Rulesets

Exclusions

Exceptions

A pre-configured rule set is enabled by default. This rule set protects your web application from common threats defined in the top-ten OWASP categories. The default rule set is managed by the Azure WAF service. Rules are updated as needed for new attack signatures. [Learn more](#)

🕒 Policy mode is set to Detection. Detection mode monitors and logs all threat alerts if logging is enabled. [Learn more](#)

📋 Assign

➕ Add exclusions

➕ Add exceptions

🔄 Refresh

📄 Export to CSV

|

✓ Enable

🕒 Disable

✎ Change action

⌵

🔍 Filter for any field

Rule group : (All) ✕

Status : (All) ✕

Action : (All) ✕

Group by Rule set ⌵

☐	>	Rule Id	Description	Action	Status	Exclusions	Rule group	Rule set
	⌵	Microsoft_DefaultRuleSet_2.1 (190)						
☐		930100	Path Traversal Attack (/../)	Anomaly score	✓ Enabled		LFI	Microsoft_DefaultRuleSet_2.1
☐		930110	Path Traversal Attack (/../)	Anomaly score	✓ Enabled		LFI	Microsoft_DefaultRuleSet_2.1
☐		930120	OS File Access Attempt	Anomaly score	✓ Enabled		LFI	Microsoft_DefaultRuleSet_2.1
☐		930130	Restricted File Access Attempt	Anomaly score	✓ Enabled		LFI	Microsoft_DefaultRuleSet_2.1
☐		931100	Possible Remote File Indusion (RFI) Attack: URL Parameter using IP Ad...	Anomaly score	✓ Enabled		RFI	Microsoft_DefaultRuleSet_2.1
☐		931110	Possible Remote File Indusion (RFI) Attack: Common RFI Vulnerable P...	Anomaly score	✓ Enabled		RFI	Microsoft_DefaultRuleSet_2.1
☐		931120	Possible Remote File Indusion (RFI) Attack: URL Payload Used w/Traili...	Anomaly score	✓ Enabled		RFI	Microsoft_DefaultRuleSet_2.1
☐		931130	Possible Remote File Indusion (RFI) Attack: Off-Domain Reference/Link	Anomaly score	✓ Enabled		RFI	Microsoft_DefaultRuleSet_2.1
☐		932100	Remote Command Execution: Unix Command Injection	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932105	Remote Command Execution: Unix Command Injection	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932110	Remote Command Execution: Windows Command Injection	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932115	Remote Command Execution: Windows Command Injection	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932120	Remote Command Execution: Windows PowerShell Command Found	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932130	Remote Command Execution: Unix Shell Expression or Confluence Vul...	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932140	Remote Command Execution: Windows FOR/IF Command Found	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1
☐		932150	Remote Command Execution: Direct Unix Command Execution	Anomaly score	✓ Enabled		RCE	Microsoft_DefaultRuleSet_2.1

Peter Hudcovic · Azure Platform Lab

113

10 - App Service and Key Vault

Prod App Service runs Python 3.12 on P0v3.

Home

 **app-sits-prod-swc**
Web App

Search

⌵ ⌵

Browse

Stop

Swap

Restart

Delete

Refresh

Download publish profile

Reset publish profile

Share to mobile

Send us your feedback

⌵

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

AI (preview)

Resource visualizer

Deployment

Settings

Performance

App Service plan

Development Tools

API

Monitoring

Automation

Support + troubleshooting

⌵ Essentials

Resource group [\(move\)](#)

Status

Location [\(move\)](#)

Subscription [\(move\)](#)

Subscription ID

Tags [\(edit\)](#)

⌵ Properties

Monitoring

Logs

Capabilities

Notifications (0)

Recommendations

 **Web app**

Name

Publishing model

Runtime Stack

Runtime status

app-sits-prod-swc

Code

Python - 3.12

Healthy

 **Domains**

Default domain

Custom domain

[app-sits-prod-swc.azurewebsites.net](#)

[Add custom domain](#)

 **Hosting**

Plan Type

Name

Operating System

Instance Count

SKU and size

App Service plan

[asp-sits-prod-swc](#)

Linux

1

Premium0V3 (P0v3) [Scale up](#)

Default domain

App Service Plan

Operating System

Health Check

[app-sits-prod-swc.azurewebsites.net](#)

[asp-sits-prod-swc \(P0v3: 1\)](#)

Linux

100.00% (Healthy 1 / Degraded 0)

 **Deployment Center**

Deployment logs

Last deployment

Deployment provider

[View logs](#)

Loading deployments...

None

 **Application Insights**

Name

Region

[appi-sits-prod-swc](#)

Sweden Central

 **Networking**

Inbound IP addresses

Private endpoint connections

Virtual network integration

51.12.74.1, 2603:1020:1004:2::507

[1 private endpoint](#)

[vnet-sits-prod-swc/snet-app](#)

Peter Hudcovic · Azure Platform Lab

115

Public access disabled; traffic only through the private endpoint.

Home > app-sits-prod-swc

 **app-sits-prod-swc** | Networking

Web App

Search

⌵ ⏪

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

AI (preview)

Resource visualizer

> Deployment

▼ Settings

{x}

 Environment variables

|||

 Configuration

☰

 Instances

👤

 Authentication

🔑

 Identity

☁

 Backups

🌐

 Custom domains

📜

 Certificates

🔄 Refresh

🛠 Troubleshoot

💬 Send us your feedback

Check your network configuration. Select any of the features listed below to change your network setup. [Learn more](#)

 **Inbound traffic configuration**

Public network access

Disabled

App assigned address

Not supported

Private endpoints

1 private endpoint

Inbound IPv4 addresses

10.30.5.7

Inbound IPv6 addresses

2603:1020:1004:2::507

Optional inbound services

Azure Front Door

[View details](#)

 **Outbound traffic configuration**

Virtual network integration

vnet-sits-prod-swc/snet-app

Hybrid connections

Not configured

Outbound DNS

Inherited (from virtual network)

Outbound IPv4 addresses

172.160.250.161, 4.165.42.206, 135.22... [Show More](#)

Outbound IPv6 addresses

2603:1020:1001:12::53c, 2603:1020:10... [Show More](#)

Integration subnet configuration

NAT gateway

ng-sits-prod-swc

Network security group

nsg-sits-prod-app-swc

User defined route

Not configured

Peter Hudcovic · Azure Platform Lab

116

Web App private endpoint provisioned.

Home

 **pe-app-sits-prod-swc**
Private endpoint

Is my private endpoint resilient

×

Search

◇

«

➔ Move

▼

 Delete

 Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

^ Essentials

Resource group [\(move\)](#)

: [rg-sits-prod-app-swc](#)

Location [\(move\)](#)

: Sweden Central

Subscription [\(move\)](#)

: [sub-app-prod](#)

Subscription ID

: 61845030-2daa-4797-a97a-59a7fd8d478f

Provisioning state

: Succeeded

Tags [\(edit\)](#)

:

costCenter : sits-lab

environment : prod

Show more (1)

[JSON View](#)

Virtual network/subnet

: [vnet-sits-prod-swc/snet-pe](#)

Network interface

: [nic-pe-app-sits-prod-swc](#)

Private link resource

: [app-sits-prod-swc](#)

Target sub-resource

: sites

Connection status

: Approved

Request/Response

:

Peter Hudcovic · Azure Platform Lab

117

App Key Vault: public access disabled.

Home > kv-sits-prod-app-swc

 kv-sits-prod-app-swc | Networking

Key vault

Search

»

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Settings

Access configuration

Networking

Microsoft Defender for Cloud

Properties

Locks

Monitoring

Automation

Help

Firewalls and virtual networks

Private endpoint connections

Allow access from:

☐ Allow public access from all networks

☐ Allow public access from specific virtual networks and IP addresses

☒ Disable public access

 No public traffic will be able to access this resource. [Learn more about key vault network security configuration](#)

Exception

Enabling access to resources requires you allow trusted Microsoft services to bypass firewall.

☐ Allow trusted Microsoft services to bypass this firewall

 This setting is related to firewall only. In order to access this key vault, the trusted service must also be given explicit permissions the [Access Policies](#) section. [configuration](#)

Peter Hudcovic · Azure Platform Lab

118

App Key Vault private endpoint connection approved.

Home > kv-sits-prod-app-swc

 kv-sits-prod-app-swc | Networking

Key vault

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Settings

Access configuration

Networking

Microsoft Defender for Cloud

Firewalls and virtual networks

Private endpoint connections

Private endpoints allow access to this resource using a private IP address form a virtual network, effectively bringing the service into your virtual network. [Learn more about private endpoint connections](#)

+ Create

↺ Refresh

|

✓ Approve

✗ Reject

🗑 Remove

Filter by name...

Connection states == All

☐ Private endpoint	Connection name	Sub-resource	Subnet	Connection state	Description
☐	pe-kv-sits-prod-app-swc	psc-kv-sits-prod-app-swc	vault	vnet-sits-prod-swc/snet-pe	Approved

Peter Hudcovic · Azure Platform Lab

119

Certificate vault: public access allowed with restrictions (known lab limit).

Home > kv-sits-prod-cert-swc

 kv-sits-prod-cert-swc | Networking

Key vault

☆ ...

Search

◊ <<

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Settings

Access configuration

Firewalls and virtual networks

Private endpoint connections

Allow access from:

☒ Allow public access from all networks

☐ Allow public access from specific virtual networks and IP addresses

☐ Disable public access

 Traffic from all public networks can access this resource. This is not recommended for private applications or environments.

Certificate vault private endpoint connection.

Home > kv-sits-prod-cert-swc

 kv-sits-prod-cert-swc | Networking

☆ ...

Key vault

🔍 < <<

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

> Objects

▼ Settings

Access configuration

Networking

Microsoft Defender for Cloud

...

Firewalls and virtual networks

Private endpoint connections

Private endpoints allow access to this resource using a private IP address form a virtual network, effectively bringing the service into your virtual network. [Learn more about private endpoint connections](#)

+ Create

🔄 Refresh

|

✓ Approve

✗ Reject

🗑 Remove

⚠ Firewall settings allow access from all networks. The security provided by private endpoints will not be satisfied.

Connection states == **All**

☐ Private endpoint	Connection name	Sub-resource	Subnet	Connection state	Description
☐ pe-kv-sits-prod-cert-swc	psc-kv-sits-prod-cert-swc	vault	vnet-sits-prod-swc/snet-pe	Approved	

Admin vault: public access allowed with restrictions (known lab limit).

Home > kv-sits-prod-adm-swc

 kv-sits-prod-adm-swc | Networking

☆ ...

Key vault

Search

⌵ ⏪

 Overview

 Activity log

 Access control (IAM)

 Tags

 Diagnose and solve problems

 Access policies

 Resource visualizer

 Events

 Objects

 Settings

 Access configuration

 **Networking**

Firewalls and virtual networks

Private endpoint connections

Allow access from:

☒ Allow public access from all networks

☐ Allow public access from specific virtual networks and IP addresses

☐ Disable public access

 Traffic from all public networks can access this resource. This is not recommended for private applicatic

Admin vault private endpoint connection.

Home > kv-sits-prod-adm-swc

 kv-sits-prod-adm-swc | Networking

Key vault

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Settings

Access configuration

Networking

Microsoft Defender for Cloud

Firewalls and virtual networks

Private endpoint connections

Private endpoints allow access to this resource using a private IP address form a virtual network, effectively bringing the service into your virtual network. [Learn more about private endp](#)

+ Create

↺ Refresh

✓ Approve

✗ Reject

🗑 Remove

⚠ Firewall settings allow access from all networks. The security provided by private endpoints will not be satisfied.

Connection states == **All**

☐ Private endpoint	Connection name	Sub-resource	Subnet	Connection state
☐ pe-kv-sits-prod-adm-swc	psc-kv-sits-prod-adm-swc	vault	vnet-sits-prod-swc/snet-pe	Approved

Peter Hudcovic · Azure Platform Lab

123

11 · Production protection and admin access

Delete lock on the Prod app resource group, removable only by the destroy pipeline.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

admin@peterhudcovicg...
DEFAULT DIRECTORY (PETERHU...

Home > rg-sits-prod-app-swc

rg-sits-prod-app-swc | Locks

Resource group

How do locks prevent accidental deletion of Azure resources?

What are the best practices for locking resources in Azure?

Can a resource locked for 'ReadOnly' be modified?

Search

+ Add

Subscription

Refresh

Feedback

Overview

Activity log

Access control (IAM)

Tags

Analyze (Preview)

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Locks

Cost Management

Monitoring

Automation

Help

Lock name	Lock type	Scope	Notes
lock-sits-prod-application	Delete	rg-sits-prod-app-swc	SITS production: removed only by the prod-destroy pipeline.

Edit

Delete

One Bastion per environment.

Home

Bastions

☆ ... 

Identify connectivity issues in bastion resources

Analyze performance of bastion resources

Track metrics for bastion hosts

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Create

 Manage view

▼

 Refresh

 Export to CSV

 Open query

|

 Assign tags

+ Add to service group

 Filter for any field...

Subscription equals **all**

Resource Group equals **all**

×

Location equals **all**

×

+ Add filter

☐ Name ↑	Type	Resource Group	Location	Subscription
☐  bas-sits-dev-swc	... Bastion	rg-sits-dev-network-swc	Sweden Central	sub-app-nonprod
☐  bas-sits-prod-swc	... Bastion	rg-sits-prod-network-swc	Sweden Central	sub-app-prod
☐  bas-sits-test-swc	... Bastion	rg-sits-test-network-swc	Sweden Central	sub-app-nonprod

Prod uses Bastion Basic in AzureBastionSubnet.

Home

 **bas-sits-prod-swc**
Bastion

 Run diagnostic for this bastion

Analyze health of this bastion

Search

Search

Delete

Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : [rg-sits-prod-network-swc](#)

Location : Sweden Central

Subscription (move) : [sub-app-prod](#)

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Virtual network/subnet : [vnet-sits-prod-swc/AzureBastionSubnet](#)

Public DNS name : bst-c9d87a95-58c8-43b3-b934-b1aceb7e3cb8.bastion.azure.com

Public IPv4 address : [-](#)

Public IPv6 address : [-](#)

Tier : Basic

Provisioning state : Succeeded

Availability zone : None

Tags (edit) : costCenter : sits-lab environment : prod owner : peter

Tutorials

Sessions

Free training from Microsoft

101

Connect to virtual machines through the Azure portal by using Azure Bastion

8 units • 40 min

Deploy Azure Bastion to securely connect to Azure virtual machines directly within the Azure portal.

[Start "Connect to virtual machines" training.](#)

101

Configure and manage virtual networks for Azure administrators

15 units • 13 hr 39 min

Learn how to configure and manage a range of Azure network capabilities including connectivity services and network monitoring services.

[Start "Configure and manage virtual networks" training.](#)

Peter Hudcovic · Azure Platform Lab

127

Prod Ops VM: no public IP.

Home

vm-sits-prod-ops-swc

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

Feedback

CLI / PS

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Windows Admin Center

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Essentials

Resource group (move) : rg-sits-prod-network-swc

Status : Running

Location : Sweden Central (Zone 1)

Subscription (move) : sub-app-prod

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Availability zone : 1

Tags (edit) : costCenter : sits-lab environment : prod owner : peter

Properties

Monitoring

Capabilities (8)

Recommendations

Tutorials

Virtual machine

Computer name : ops-prod

Operating system : Windows (Windows Server 2022 Datacenter Azure Edition)

VM generation : V2

VM architecture : x64

Agent status : Ready

Agent version : 2.7.41491.1243

Hibernation : Disabled

Host group : -

Host : -

Proximity placement group : -

Colocation status : N/A

Capacity reservation group : -

Disk controller type : NVMe

Azure Spot

Azure Spot : -

Azure Spot eviction policy : -

Availability + scaling

Availability zone (edit) : 1

Extended zone : -

Availability set : -

Scale Set (attach) : -

Security

Security type : Trusted launch

Enable secure boot : Enabled

Enable vTPM : Enabled

Integrity monitoring : Disabled

Metadata Security Protocol : Disabled

WireServer : Disabled

Networking

Public IP address : 172.160.250.161 (NAT gateway ng-sits-prod-swc)
1 associated public IPs

Public IP address (IPv6) : -

Private IP address : 10.30.1.4

Private IP address (IPv6) : -

Virtual network/subnet : vnet-sits-prod-swc/snet-ops

DNS name : -

Size

Size : Standard D2s v6

vCPUs : 2

RAM : 8 GiB

Threads per core : 2

Source image details

Source image publisher : MicrosoftWindowsServer

Source image offer : WindowsServer

Source image plan : 2022-datacenter-azure-edition

Disk

OS disk : osdisk-vm-sits-prod-ops-swc

Encryption at host : Disabled

Azure disk encryption : Not enabled

Ephemeral OS disk : N/A

Data disks : 0

Auto-shutdown

Auto-shutdown : Not enabled

Scheduled shutdown : -

Peter Hudcovic · Azure Platform Lab

128

Connect to the Prod VM through Bastion.

Home > vm-sits-prod-ops-swc

 **vm-sits-prod-ops-swc** | Connect

Virtual machine

◊ <<

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

 **Connect**

 Bastion

 Windows Admin Center

> Networking

Settings

 Disks

 Extensions + applications

Now view, configure, and even save your connection settings — all in one place. Have comments or s

Refresh

Reset password or keys

Manage JIT

Troubleshoot

Feedback

CONFIGURED

AZURE PORTAL

 **Bastion (Basic, Standard, or Premium)**

Advanced capabilities including custom inbound port, RDP on desktop, and host scaling.
[Learn more](#)

Go to Bastion

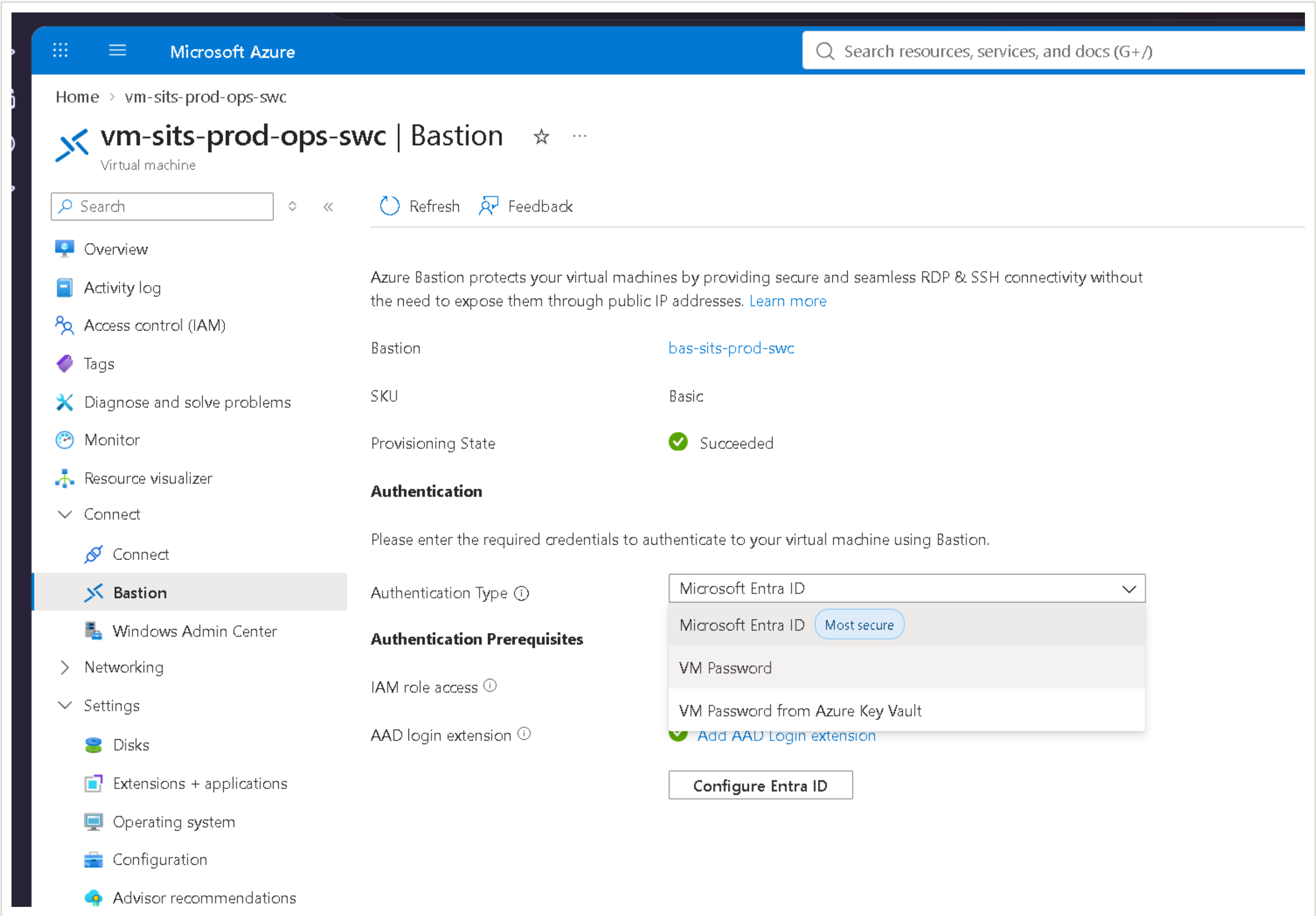


< More ways to connect (4)

Peter Hudcovic · Azure Platform Lab

129

Bastion offers Entra ID or VM password sign-in.



VM password sign-in, kept for emergency access.

Home > vm-sits-prod-ops-sw

vm-sits-prod-ops-sw | Bastion

Virtual machine

Search

RefreshFeedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Bastion

Windows Admin Center

> Networking

> Settings

Disks

Extensions + applications

Azure Bastion protects your virtual machines by providing secure and seamless RDP & SSH connectivity without the need to expose them through public IP addresses. [Learn more](#)

Bastionbas-sits-prod-sw

SKUBasic

Provisioning StateSucceeded

Authentication

Please enter the required credentials to authenticate to your virtual machine using Bastion.

Authentication Type ⓘVM Password

Username * ⓘopsadmin

VM Password * ⓘ

> Connection Settings

Entra ID login extension installed on the Prod VM.

Home > vm-sits-prod-ops-swc

vm-sits-prod-ops-swc | Extensions + applications

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Windows Admin Center

Networking

Settings

Disks

Extensions + applications

Operating system

Extensions

VM Applications

+ Add

Refresh

Update

Enable automatic upgrade

Disable automatic upgrade

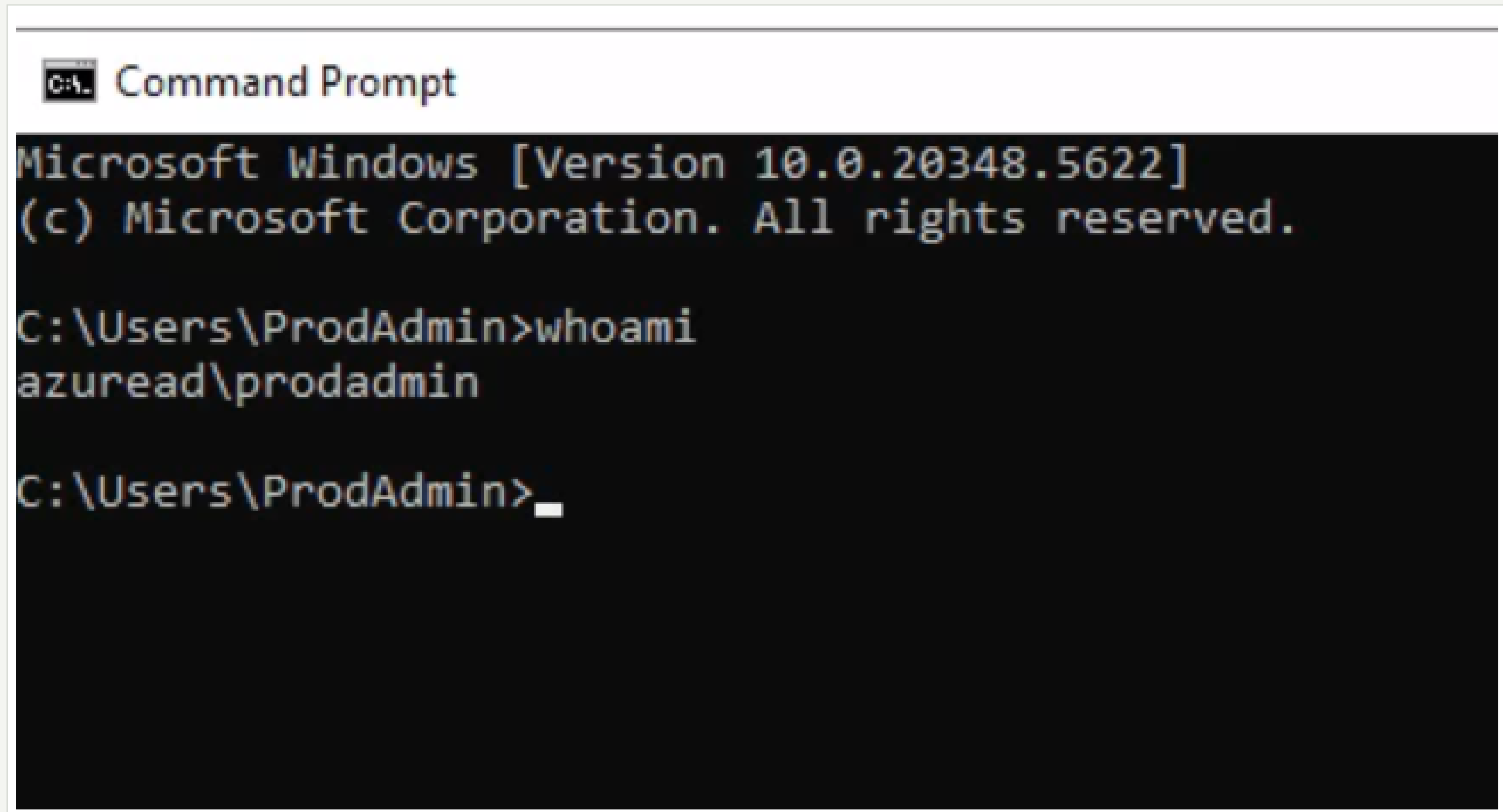
Feedback

Search to filter items...

Showing all 1 items

	Name	Type	Version	Latest Version	Status	Automatic upgrade status
	AADLoginForWindows	Microsoft.Azure.ActiveDirectory.AADLoginForWindows	2.2.0.0	2.2.0.0	Provisioning succeeded	Disabled

Entra ID sign-in: whoami returns azuread\prodadmin.

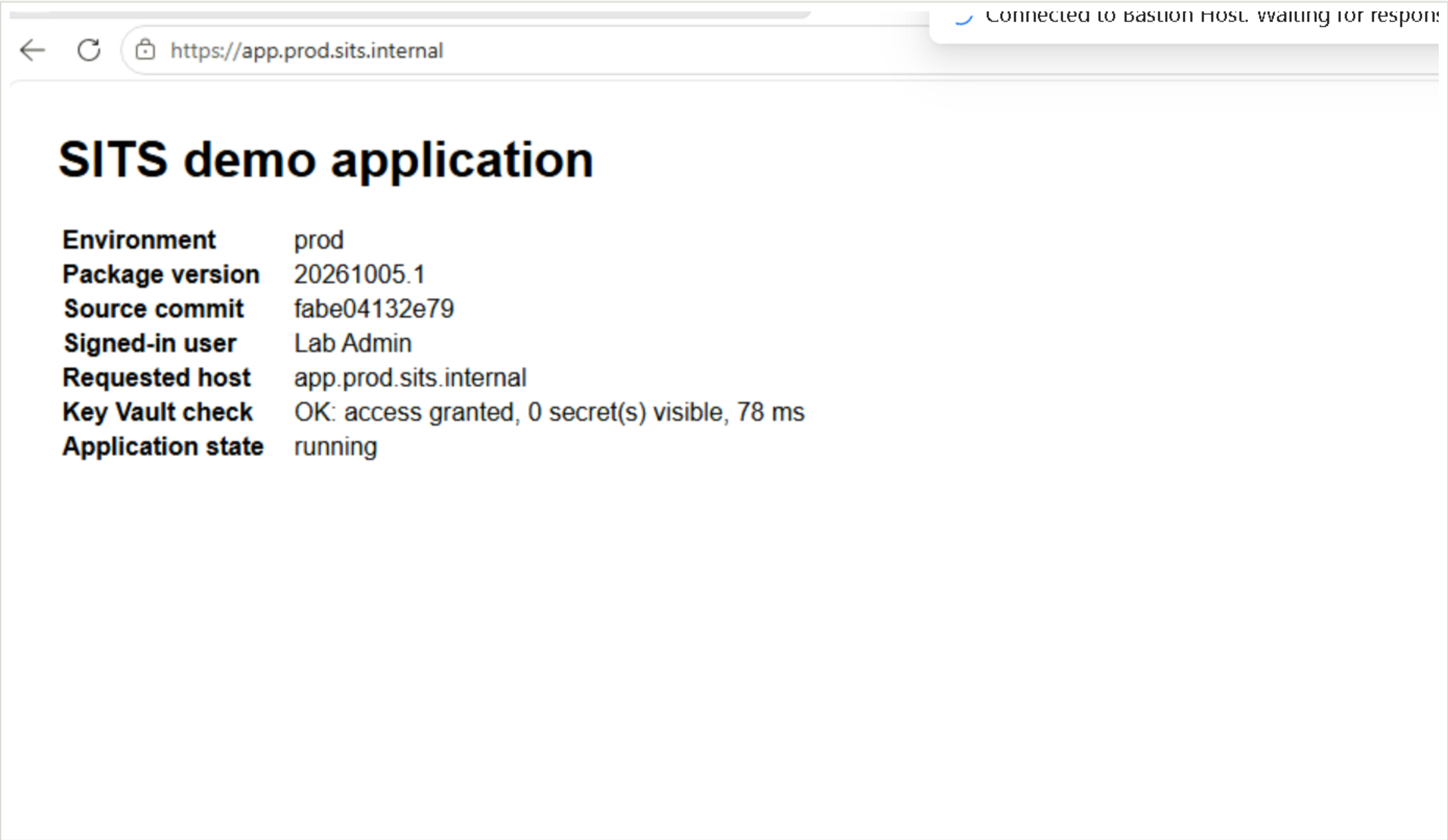


The screenshot shows a Windows Command Prompt window titled "C:_ Command Prompt". The window displays the following text:

```
Microsoft Windows [Version 10.0.20348.5622]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\ProdAdmin>whoami  
azuread\prodadmin  
  
C:\Users\ProdAdmin>_
```

12 · Application end to end

Prod app over trusted HTTPS: same package and commit, Entra sign-in, Key Vault OK.



13 · Monitoring and alerts

Four alert rules per environment, all enabled.

Home > Monitor | Alerts

Alert rules

+ Create

Columns

Refresh

Export to CSV

Open query

Delete

Enable

Disable

Create an autonomous observability agent: Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#)

Create

Search

Subscription : 2 selected

Target scope : all

Target resource type : all

Signal type : all

Severity : all

Status : Enabled

Add tag filter

No grouping

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ alert-sits-dev-app-unhealthy	HealthCheckStatus < 100	1 - Error	app-sits-dev-swc	App Service	Metrics	Enabled
☐ alert-sits-dev-gateway-backend-unheal...	UnhealthyHostCount > 0	1 - Error	agw-sits-dev-swc	Application gateway	Metrics	Enabled
☐ alert-sits-dev-keyvault-access-denied	Table rows > 0	2 - Warning	log-sits-dev-swc	Log Analytics workspace	Log search	Enabled
☐ alert-sits-dev-waf-request-matched	Table rows > 0	3 - Informational	log-sits-dev-swc	Log Analytics workspace	Log search	Enabled
☐ alert-sits-prod-app-unhealthy	HealthCheckStatus < 100	1 - Error	app-sits-prod-swc	App Service	Metrics	Enabled
☐ alert-sits-prod-gateway-backend-unhe...	UnhealthyHostCount > 0	1 - Error	agw-sits-prod-swc	Application gateway	Metrics	Enabled
☐ alert-sits-prod-keyvault-access-denied	Table rows > 0	2 - Warning	log-sits-prod-swc	Log Analytics workspace	Log search	Enabled
☐ alert-sits-prod-waf-request-matched	Table rows > 0	3 - Informational	log-sits-prod-swc	Log Analytics workspace	Log search	Enabled
☐ alert-sits-test-app-unhealthy	HealthCheckStatus < 100	1 - Error	app-sits-test-swc	App Service	Metrics	Enabled
☐ alert-sits-test-gateway-backend-unheal...	UnhealthyHostCount > 0	1 - Error	agw-sits-test-swc	Application gateway	Metrics	Enabled
☐ alert-sits-test-keyvault-access-denied	Table rows > 0	2 - Warning	log-sits-dev-swc	Log Analytics workspace	Log search	Enabled
☐ alert-sits-test-waf-request-matched	Table rows > 0	3 - Informational	log-sits-dev-swc	Log Analytics workspace	Log search	Enabled
☐ Failure Anomalies - appi-sits-prod-swc	Failure Anomalies detected	3 - Informational	appi-sits-prod-swc	Application Insights	Smart detector	Enabled

Prod alert rules including Failure Anomalies.

Home > Monitor | Alerts

Alert rules ...

+ Create

 |

≡ Columns

↺ Refresh

↓ Export to CSV

🔗 Open query

 |

🗑 Delete

▶ Enable

☐ Disable

🔧

Create an autonomous observability agent:

Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#)

🔍 Search

Subscription : sub-app-prod

Target scope : all

Target resource type : all

Signal type : all

Severity : all

Status : Enabled

+🔍 Add tag filter

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ alert-sits-prod-app-unhealthy	HealthCheckStatus < 100	1 - Error	app-sits-prod-swc	App Service	Metrics	✔ Enabled
☐ alert-sits-prod-gateway-backend-unhe...	UnhealthyHostCount > 0	1 - Error	agw-sits-prod-swc	Application gateway	Metrics	✔ Enabled
☐ alert-sits-prod-keyvault-access-denied	Table rows > 0	2 - Warning	log-sits-prod-swc	Log Analytics workspace	Log search	✔ Enabled
☐ alert-sits-prod-waf-request-matched	Table rows > 0	3 - Informational	log-sits-prod-swc	Log Analytics workspace	Log search	✔ Enabled
☐ Failure Anomalies - appi-sits-prod-swc	Failure Anomalies detected	3 - Informational	appi-sits-prod-swc	Application Insights	Smart detector	✔ Enabled

Application Insights Failure Anomalies detector in Prod.

Home > appi-sits-prod-swc | Alerts

Alert rules

...

+ Create

|

≡ Columns

↺ Refresh

⬇ Export to CSV

🔗 Open query

|

🗑 Delete

▶ Enable

☐ Disable

🔗

Create an autonomous observability agent:

Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#)

🔍 Search

Subscription : 2 selected

Target resource type : all

Target scope : appi-sits-prod-swc

Signal type : all

Severity : all

Status : Enabled

+🔍 Add tag filter

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ Failure Anomalies - appi-sits-prod-swc	Failure Anomalies detected	3 - Informational	appi-sits-prod-swc	Application Insights	Smart detector	✔ Enabled

Peter Hudcovic · Azure Platform Lab

139

Fired alerts: Key Vault access denied and WAF matches.

Home > Monitor

Monitor

Alerts

Microsoft

Search

×

«

View as timeline (preview)

+

Create

▼

Alert rules

Action groups

Alert processing rules

Prometheus rule groups

✓

Change user response

⚡

Actions

▼

Refresh

Overview

Activity log

Alerts

Metrics

Observability agents (preview)

Issues

Logs

Health models

Change Analysis

Service health

Workbooks

Dashboards with Grafana

> Insights

> Managed Services

> Settings

> Support + Troubleshooting

Create an autonomous observability agent: Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#)

Create a...

Search

Subscription : 2 selected

Time range : Past 24 hours

Alert condition : Fired

Severity : all

Add filter

Total alerts

Critical

Error

Warning

Informational

Verbose

No grouping

Name ↑↓	Severity ↑↓	Affected resource ↑↓	Alert condition ↑↓	User response ↑↓	Fire time ↑↓
☐ alert-sits-prod-waf-request-matched	3 - Informational	log-sits-prod-swc	Fired	New	10/6/2026, 1:12 AM
☐ alert-sits-dev-waf-request-matched	3 - Informational	log-sits-dev-swc	Fired	New	10/6/2026, 1:09 AM
☐ alert-sits-test-waf-request-matched	3 - Informational	log-sits-dev-swc	Fired	New	10/6/2026, 1:08 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/6/2026, 12:59 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/6/2026, 12:49 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/5/2026, 11:34 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:27 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:22 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:22 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:21 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:06 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:46 PM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/5/2026, 10:44 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:42 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:31 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:26 PM

Prod Log Analytics workspace.

Home

 **log-sits-prod-swc**
Log Analytics workspace



How do I troubleshoot issues with this Log Analytics workspace?

What is exports failed (preview) in this Log Analytics workspace?

Show me bytes exported metrics for this Log Analytics workspace.

Search

◊

«

 Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Resource visualizer

Settings

Classic

Monitoring

Automation

Help

 Azure Monitor will enforce TLS versions 1.2 and above starting March 1, 2026. To avoid data loss and service disruption please **upgrade to AMA** and ensure your **OS supports TLS 1.2** or higher. [Learn More About TLS](#), [Learn More About Agents](#).

Essentials

Resource group [\(move\)](#) : [rg-sits-prod-foundation-swc](#)

Status : Active

Location : Sweden Central

Subscription [\(move\)](#) : [sub-app-prod](#)

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags [\(edit\)](#) :

costCenter : sits-lab

environment : prod

owner : peter

Workspace Name : log-sits-prod-swc

Workspace ID : 3efb8630-1b49-47d8-bac1-dfe3ba525f85

Pricing tier : Pay-as-you-go

Access control mode : Use resource or workspace permissions

Operational issues :  [OK](#)

Get Started

Recommendations

Get started with Log Analytics

Gateway diagnostics are sent to Log Analytics.

Home > agw-sits-prod-swc

 **agw-sits-prod-swc** | Diagnostic settings

Application gateway

Search

Refresh

Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

- Alerts
- Metrics
- Diagnostic settings**
- Logs
- Advisor recommendations

Can you explain how to send diagnostic logs to Azure Monitor?

How do diagnostic settings differ from other Azure servi

Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five differe

[Learn more about diagnostic settings](#)

Diagnostic settings

Name	Storage account	Event hub	Log Analytics workspace
diag-log-sits-prod-swc	-	-	log-sits-prod-swc

[+ Add diagnostic setting](#)

Click 'Add Diagnostic setting' above to configure the collection of the following data:

- Application Gateway Access Log
- Application Gateway Performance Log
- Application Gateway Firewall Log
- AllMetrics

Gateway diagnostics include all logs and metrics.

Home > agw-sits-prod-swc | Diagnostic settings

Diagnostic setting

...

Save

Discard

Delete

Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting name

diag-log-sits-prod-swc

Logs

Category groups ⓘ

allLogs

Categories

Application Gateway Access Log

Application Gateway Performance Log

Application Gateway Firewall Log

Metrics

AllMetrics

Destination details

Send to Log Analytics workspace

Subscription

sub-app-prod

Log Analytics workspace

log-sits-prod-swc (swedencentral)

Destination table ⓘ

Azure diagnostics

Resource specific

Archive to a storage account

Stream to an event hub

Send to partner solution

Peter Hudcovic · Azure Platform Lab

143

Prod Application Insights overview (connection string hidden).

Home

 **appi-sits-prod-swc**
Application Insights

Configure alerts based on specific conditions or thresholds in your application performance data.



Search

Search

Application Dashboard

Getting started

Search

Logs

Monitor resource group

Feedback

Favorites

Rename

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Investigate

Monitoring

Usage

Configure

Settings

Automation

Help

Essentials

Resource group (move) : [rg-sits-prod-foundation-swc](#)

Location : Sweden Central

Subscription (move) : [sub-app-prod](#)

Subscription ID : 61845030-2daa-4797-a97a-59a7fd8d478f

Tags (edit) :

costCenter : sits-lab

environment : prod

owner : peter

Instrumentation key :

Connection string :

Logs workspace : [log-sits-prod-swc](#)

OTLP connection info : [Turn on OTLP support](#)

JSON View

Show data for last:

30 minutes

1 hour

6 hours

12 hours

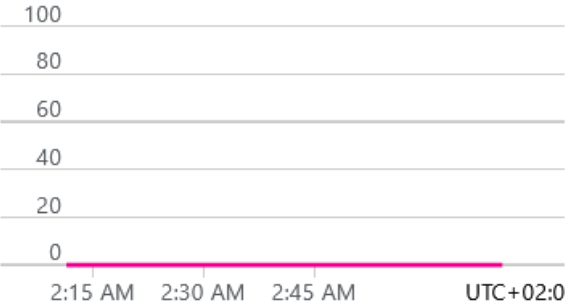
1 day

3 days

7 days

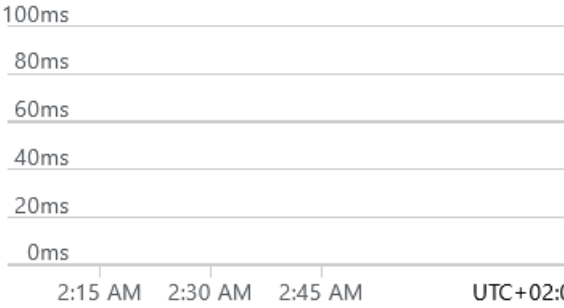
30 days

Failed requests



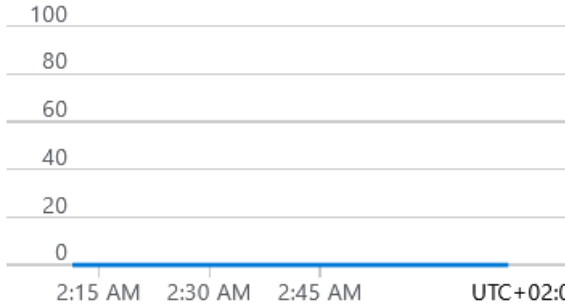
Failed requests (Count), appi-sits-prod... | 0

Server response time



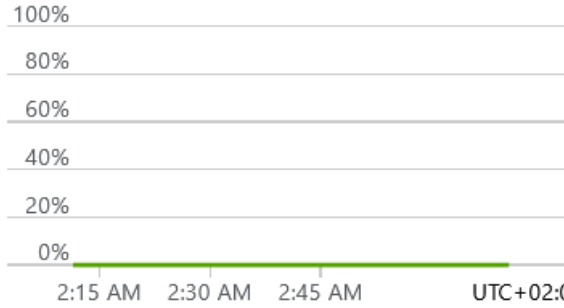
Server response time (Avg), appi-sits-pro... | --

Server requests



Server requests (Count), appi-sits-pro... | 0

Availability



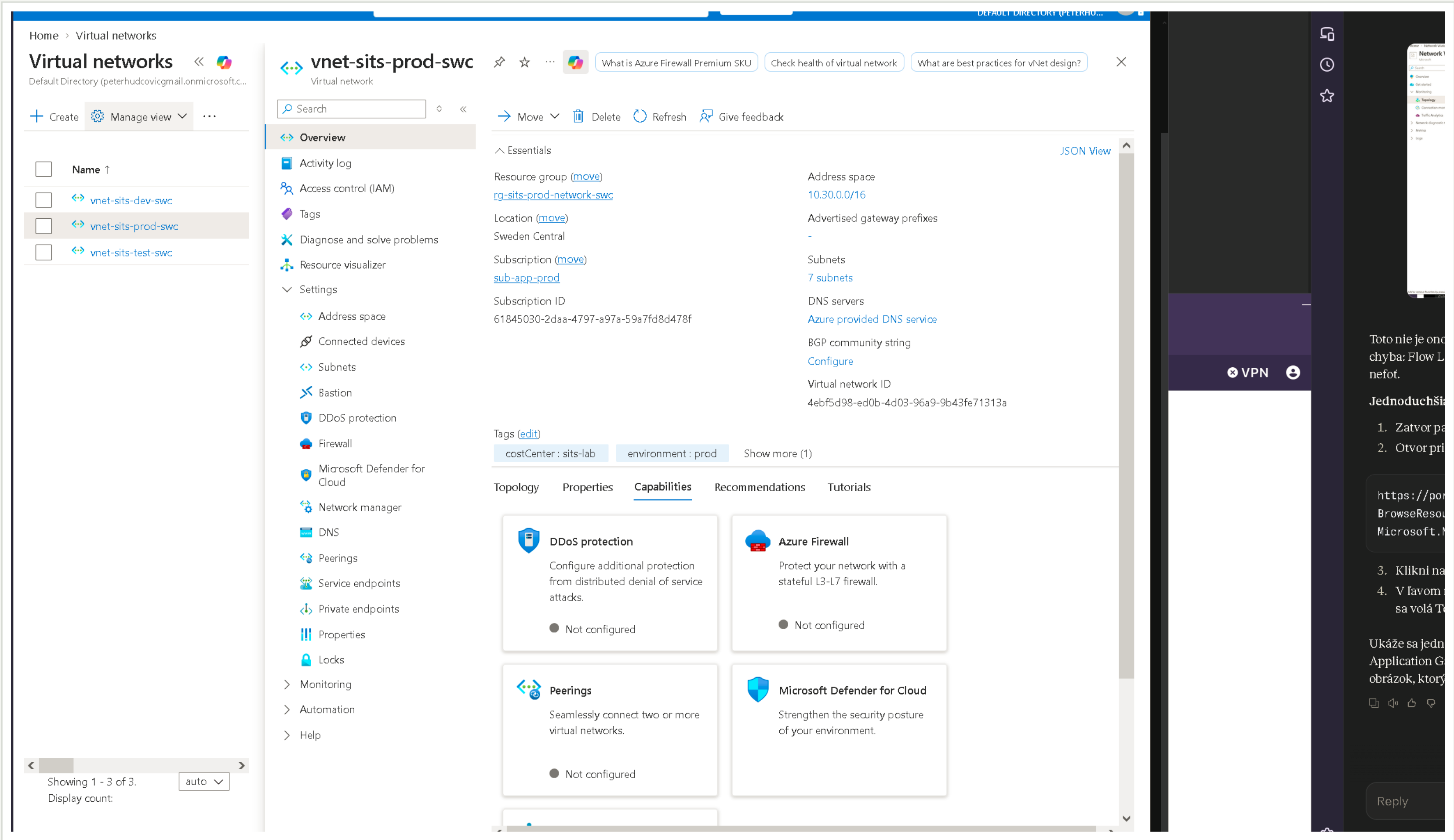
Availability (Avg), appi-sits-prod-swc | --

Peter Hudcovic · Azure Platform Lab

144

14 · Final snapshot before destroy

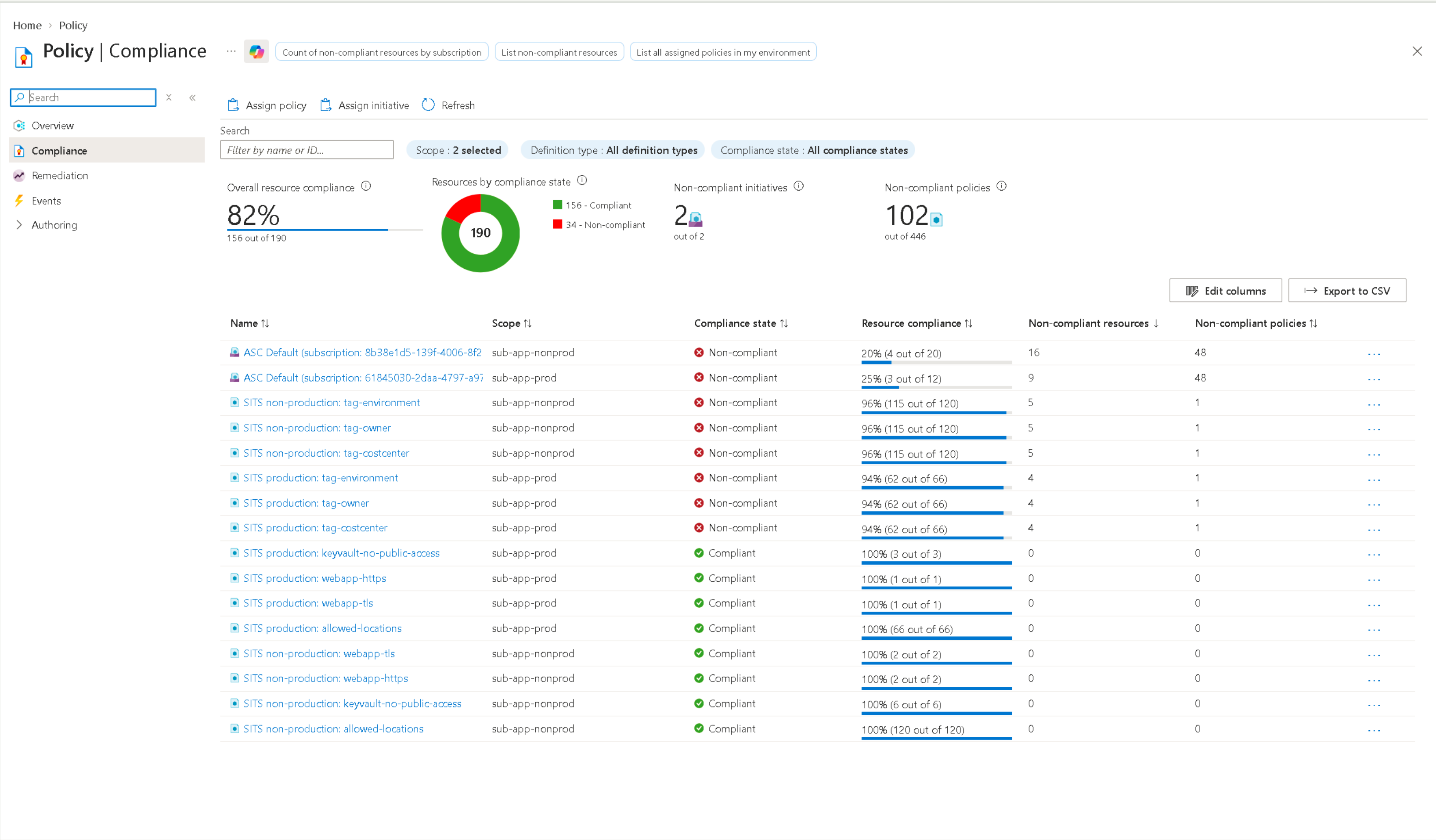
Prod VNet 10.30.0.0/16 with 7 subnets; no peering configured.



Prod VNet topology: subnets and their network security groups.

[illegible]

Azure Policy: HTTPS, TLS, Key Vault and location rules compliant; tag rules flag missing tags.



One Application Gateway with WAF per environment.

Home

Application gateways

☆ ...

Recommend connectivity for listed gateways

Execute health checks for application gateways

Review metrics for Application Gateways

+ Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Filter for any field...

Subscription equals all

Type equals all

Resource Group equals all

Location equals all

Add filter

☐	Name ↑	Type	Resource Group	Location	Subscription
☐	agw-sits-dev-swc	... Application gateway	rg-sits-dev-app-swc	Sweden Central	sub-app-nonprod
☐	agw-sits-prod-swc	... Application gateway	rg-sits-prod-app-swc	Sweden Central	sub-app-prod
☐	agw-sits-test-swc	... Application gateway	rg-sits-test-app-swc	Sweden Central	sub-app-nonprod

Log Analytics: Dev workspace (shared with Test) and a separate Prod workspace.

Home

Log Analytics workspaces

☆ ... 

List Log Analytics workspaces by Workspace ID using ARG.

How can I filter Log Analytics workspaces by pricing tier using ARG?

What are the different pricing tiers available for Log Analytics workspaces?

✕

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Create

 Open recycle bin

 Manage view ▾

 Refresh

 Export to CSV

 Open query

|

 Assign tags

+ Add to service group

 Group by none ▾

 Filter for any field...

Subscription equals **all**

Resource Group equals **all** ✕

Location equals **all** ✕

+ Add filter

☐	Name ↑		Resource Group	Location	Subscription
☐	 log-sits-dev-swc	...	rg-sits-dev-foundation-swc	Sweden Central	sub-app-nonprod
☐	 log-sits-prod-swc	...	rg-sits-prod-foundation-swc	Sweden Central	sub-app-prod

All resource groups in both subscriptions.

Home

Resource groups

...



How to manage changes with deployment tools?

Export resource groups using Bicep or Terraform

Summarize my costs by service

Default Directory (peterhudcovicgmail.onmicrosoft.com)

+ Create

 Manage view

▼

 Refresh

 Export to CSV

 Open query

|

 Assign tags

 Analyze

 Add to service group

 Filter for any field...

Subscription equals all

Location equals all

×

+ Add filter